

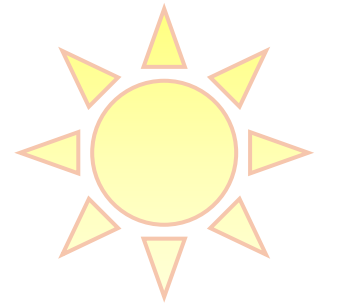
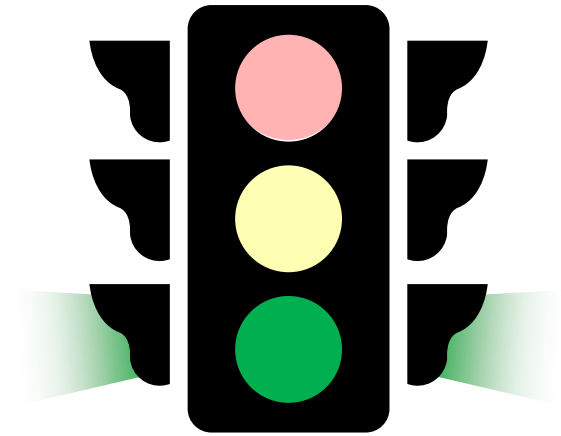


VP Ops & Eng

Influence of Braking Systems on Rail Wear and Derailment Potential

DALLAS, TX

Train Consist – Single Body

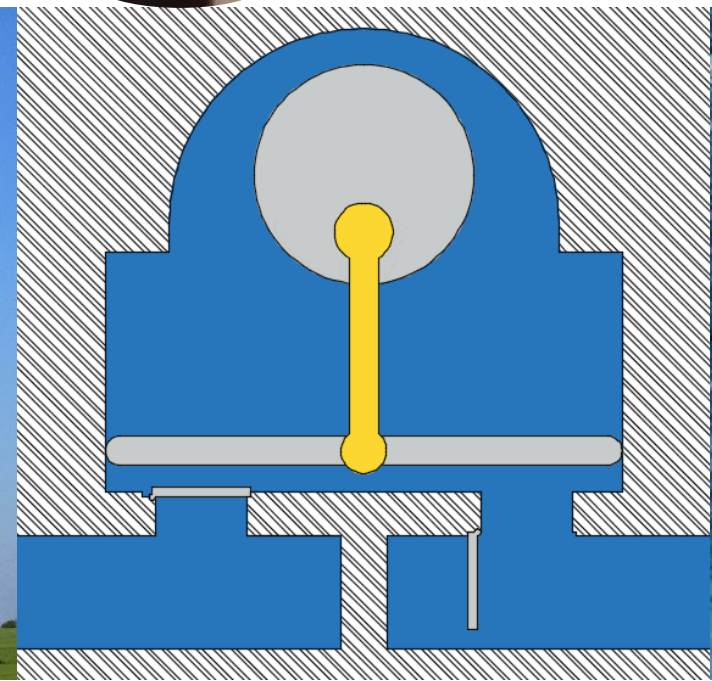


Train Consist – Individual Bodies





Locomotive Power



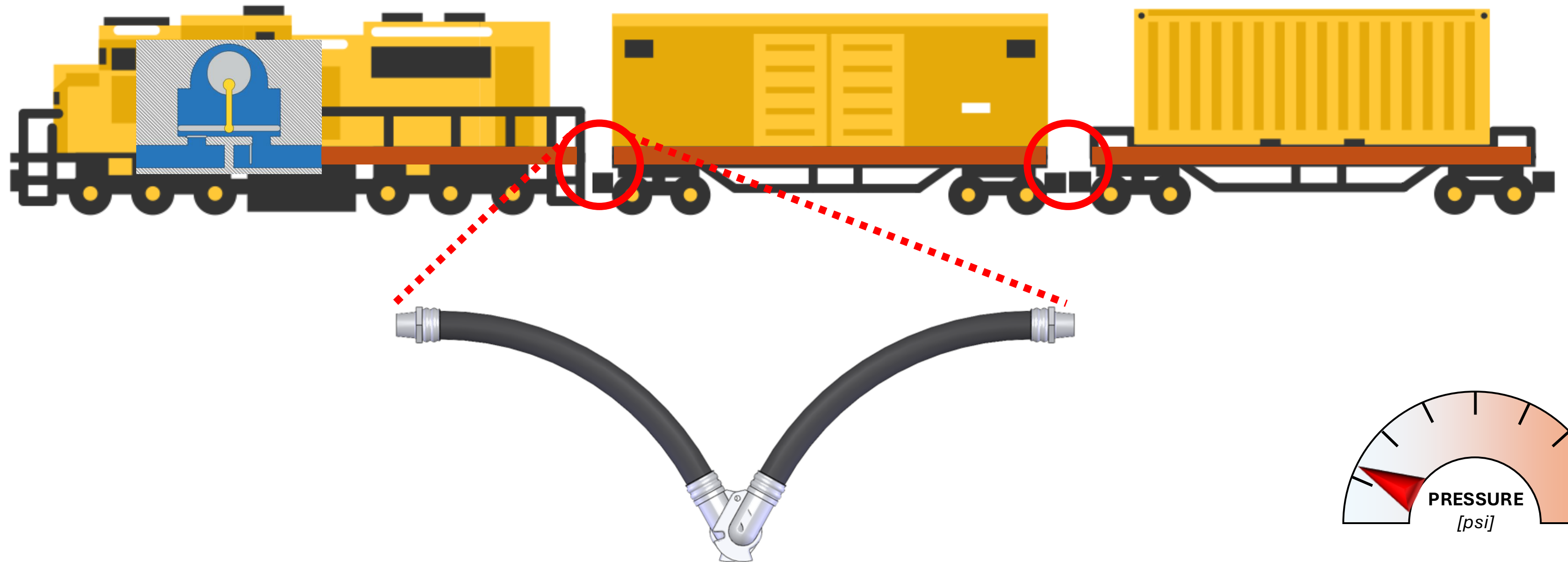
DALLAS, TX



Charging – Train Line



- Wordle:
- ✓ Train Line
 - ✓ End Hose
 - ✓ Equalization



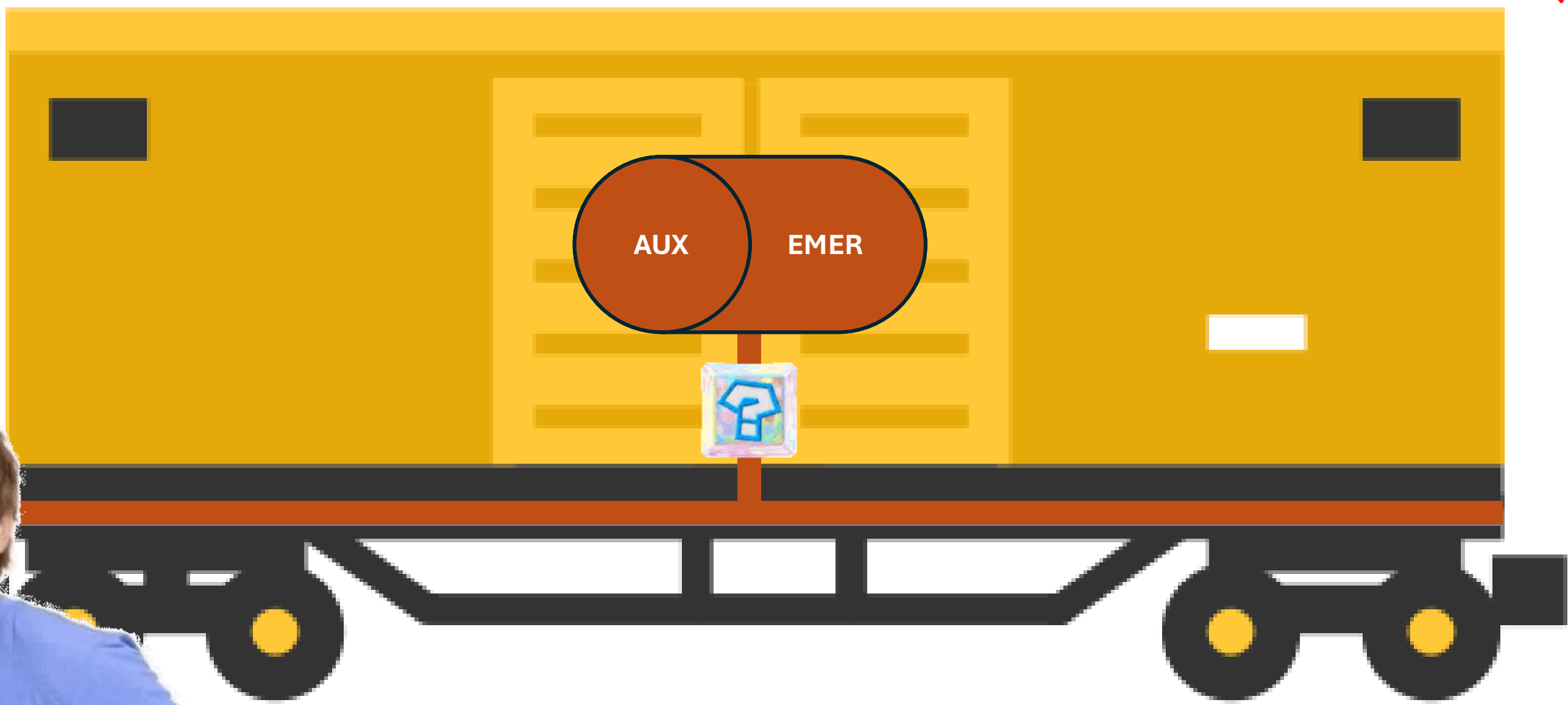
DALLAS, TX



Charging – Reservoirs



- Wordle:
- ✓ Control Valve
 - ✓ Auxiliary
 - ✓ Emergency

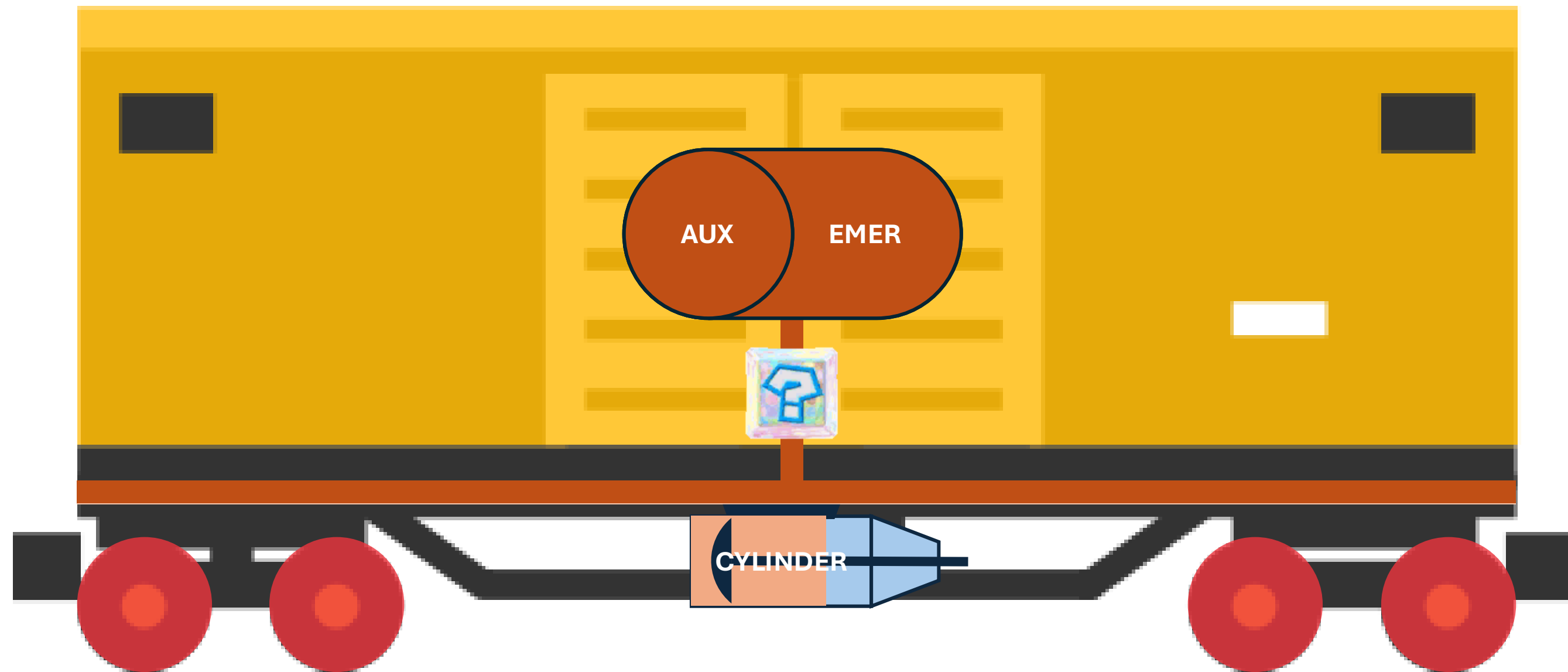




Brake Application

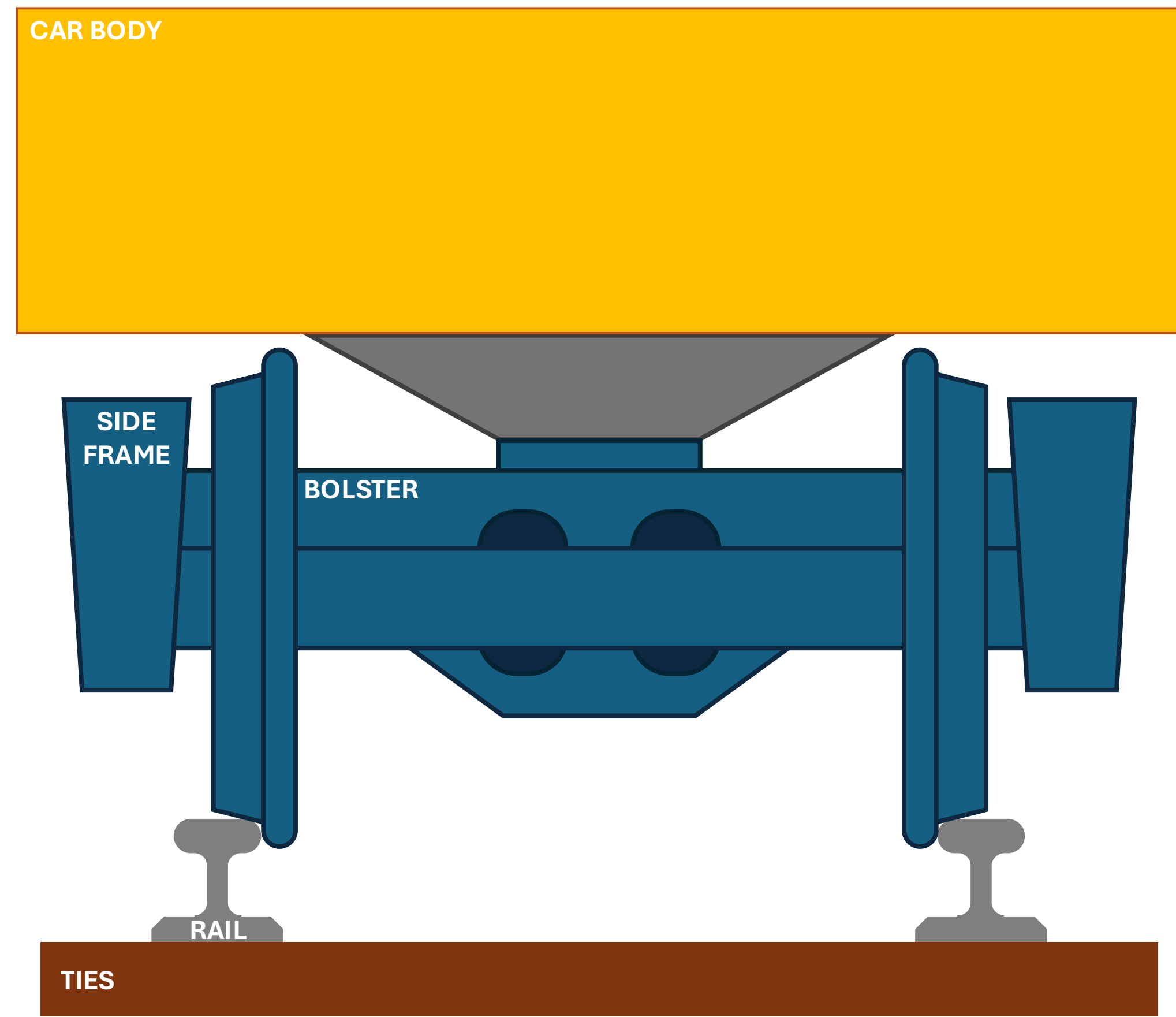


Wordle:
✓ Cylinder
✓ Reduction
✓ Equalization



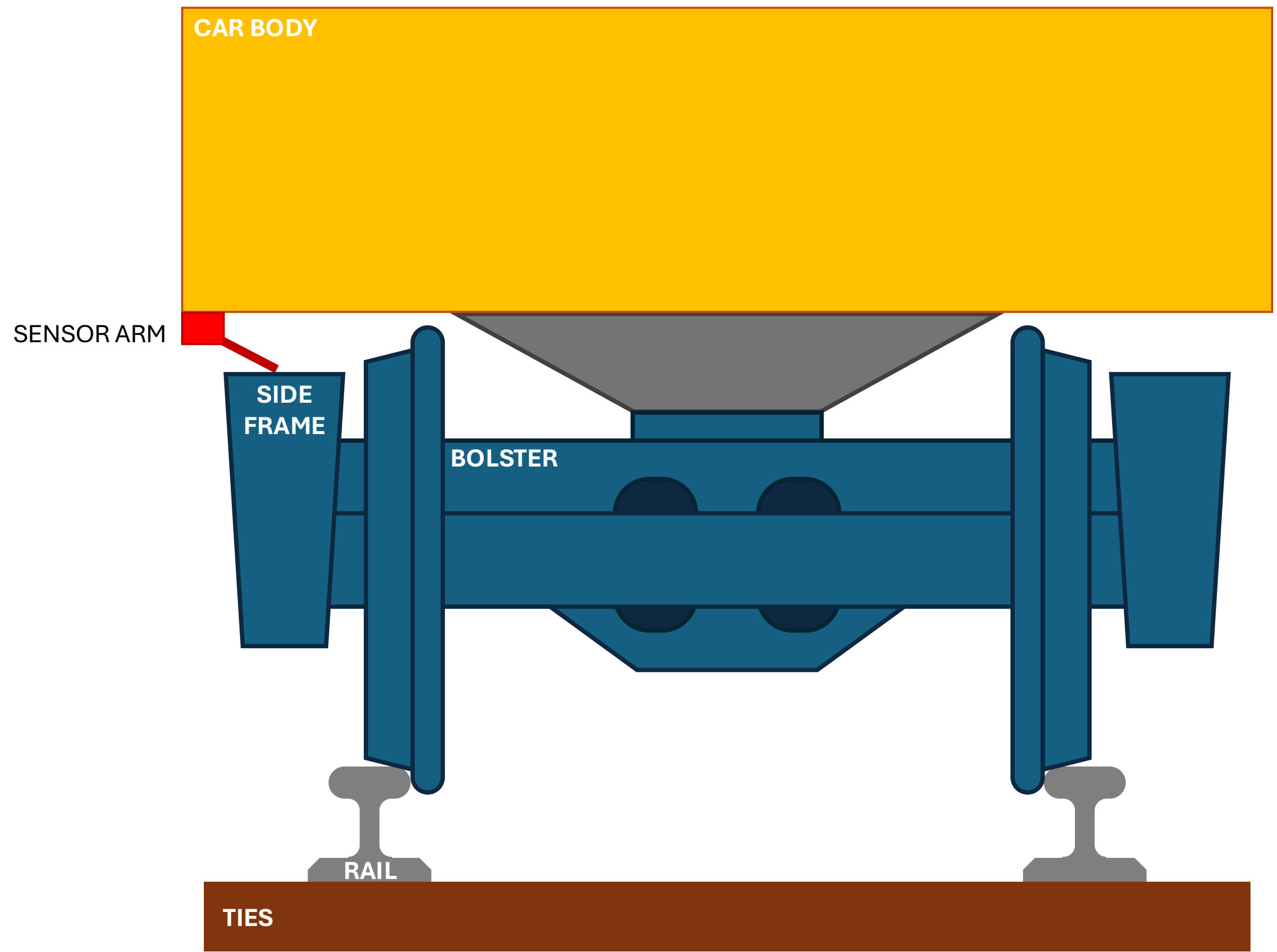


Car Body Suspension



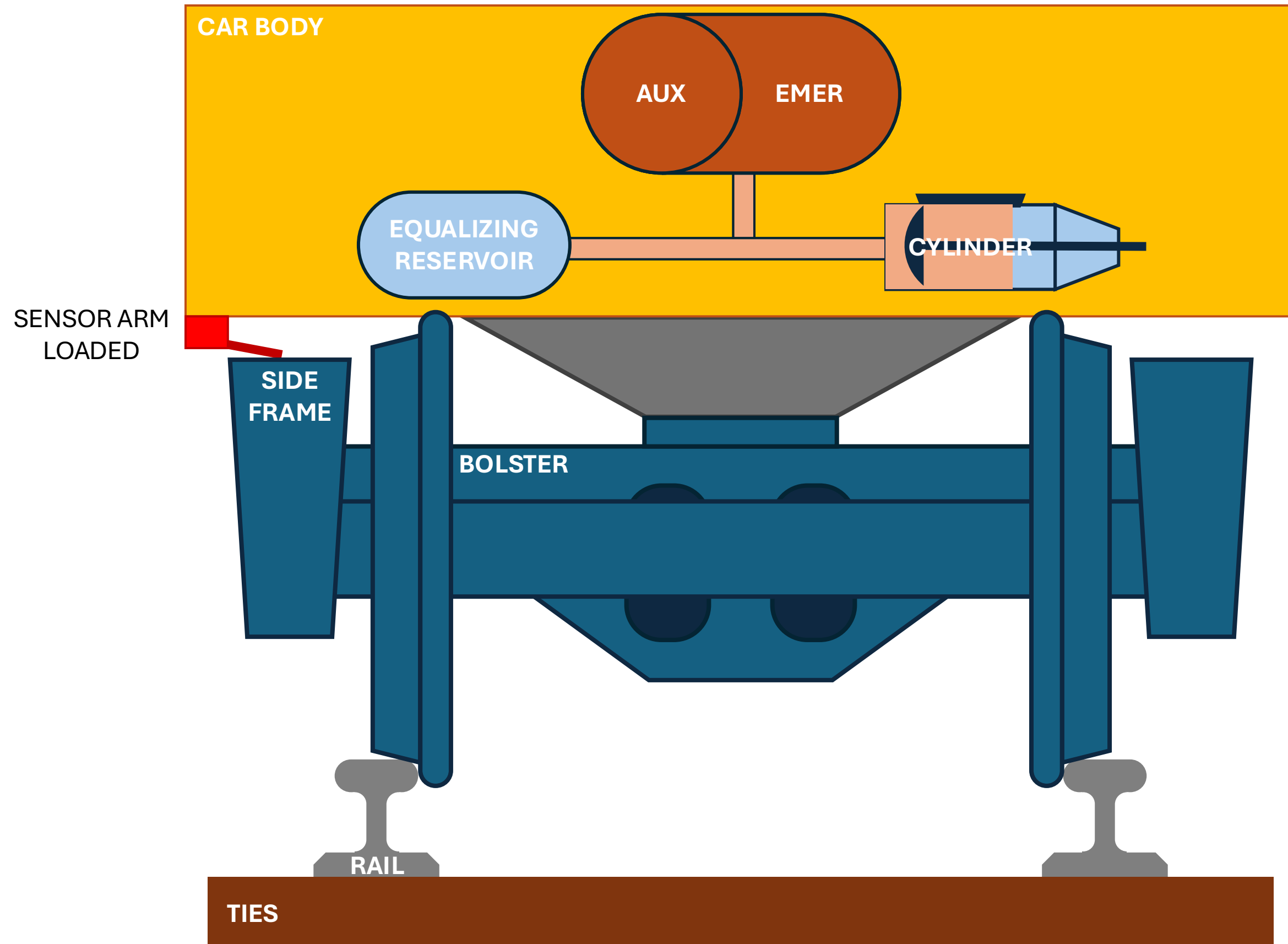


Empty-Load Devices



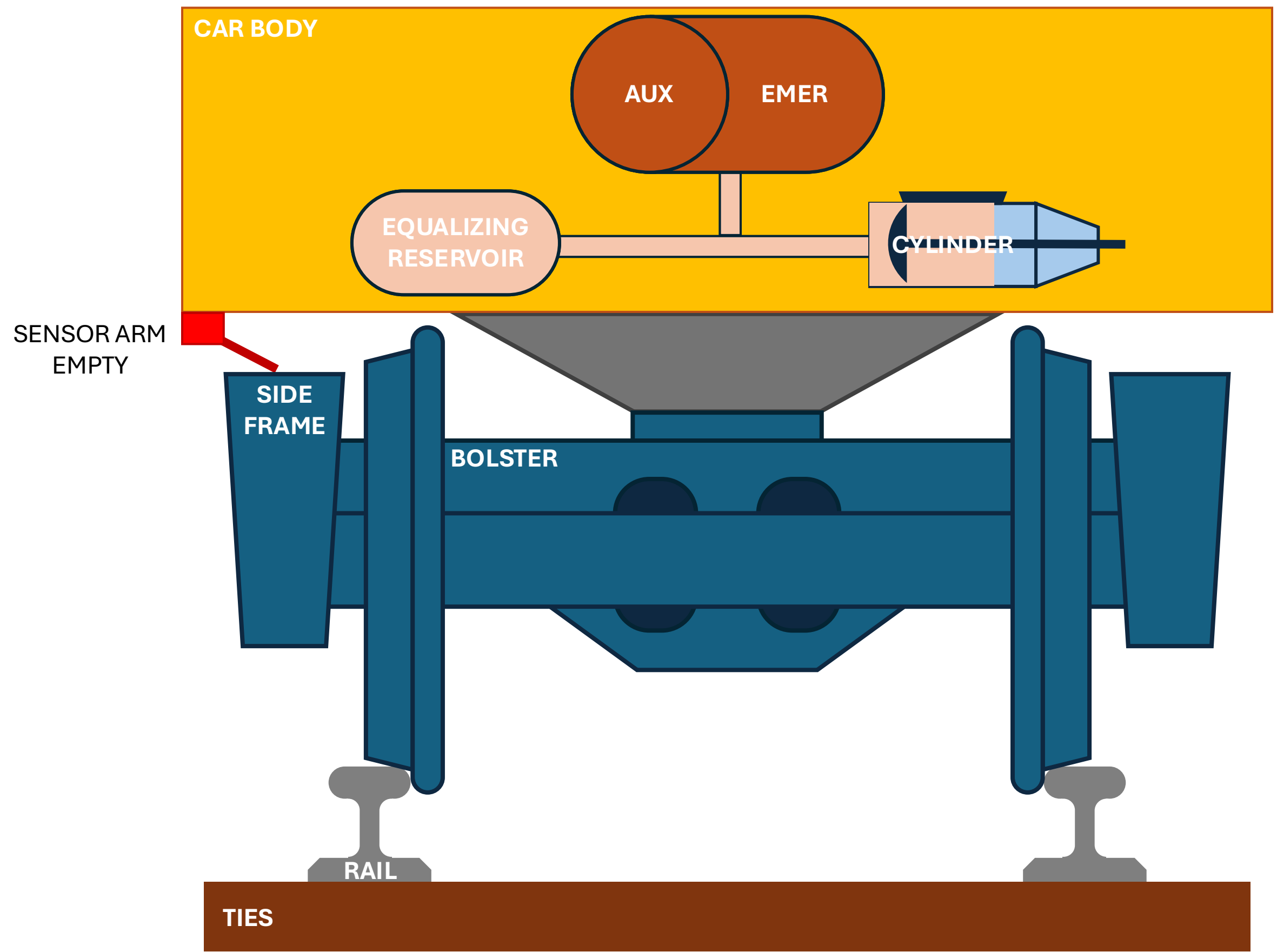


Empty-Load Devices





Empty-Load Devices

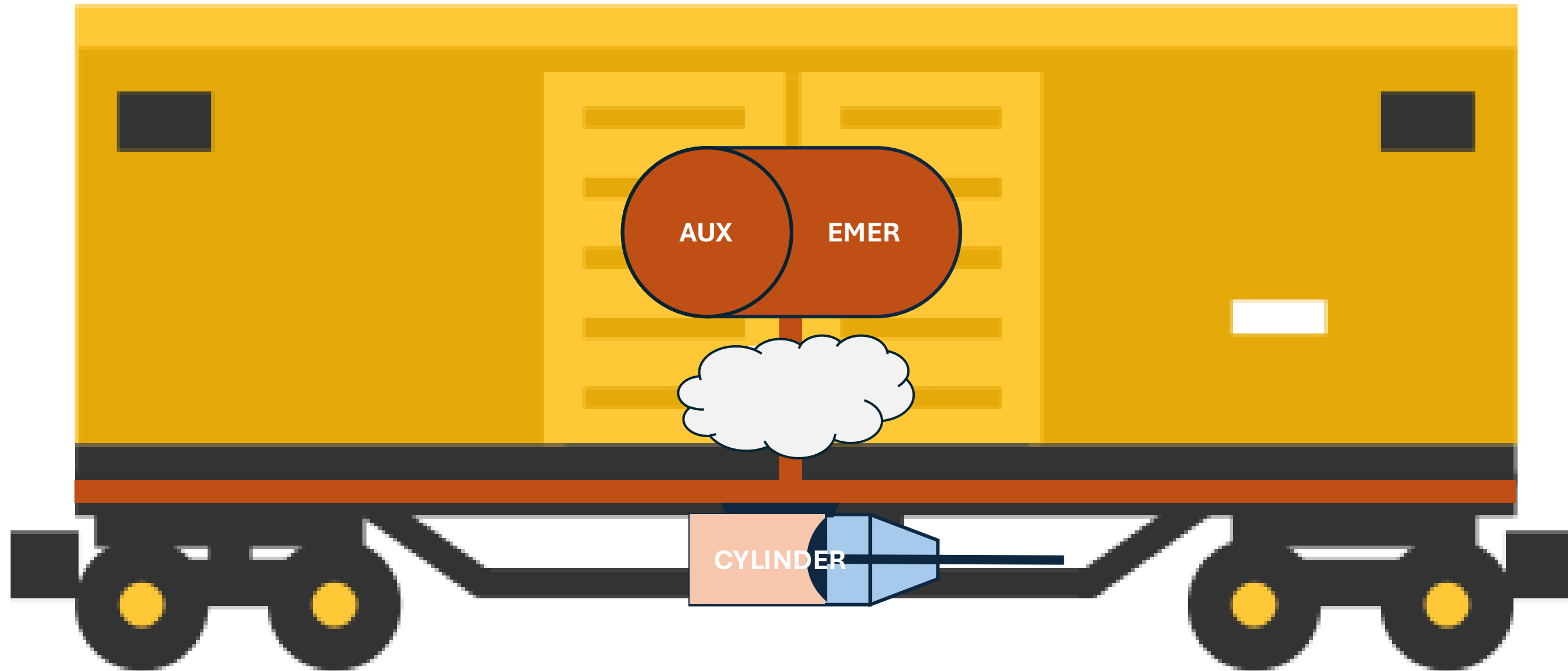




Brake Release



- Wordle:
- ✓ Release
 - ✓ Vent Valve
 - ✓ Recharge

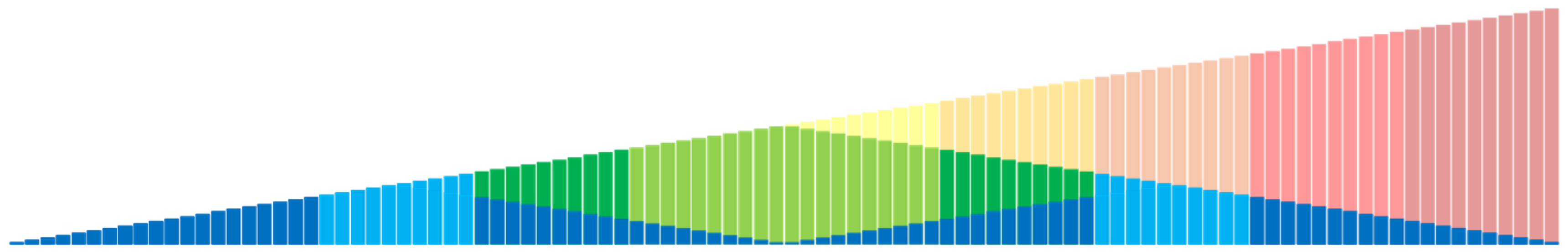




Signal Propagation



- Wordle:
- ✓ 1,125 ft/s
 - ✓ End-of-Train Device
 - ✓ Distributed power



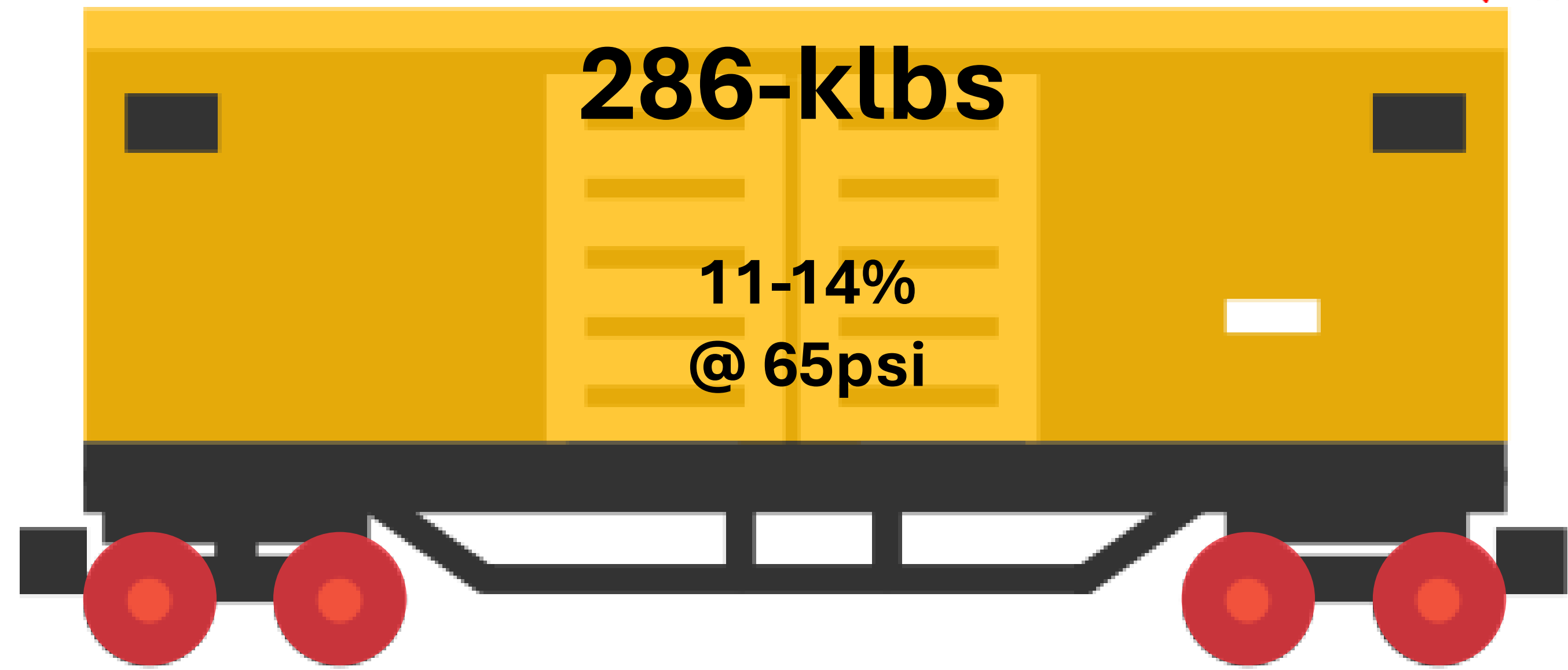
DALLAS, TX



Net Brake Ratio



- Wordle:
- ✓ Gross Rail Load
 - ✓ Brake Ratio
 - ✓ Friction Coefficient



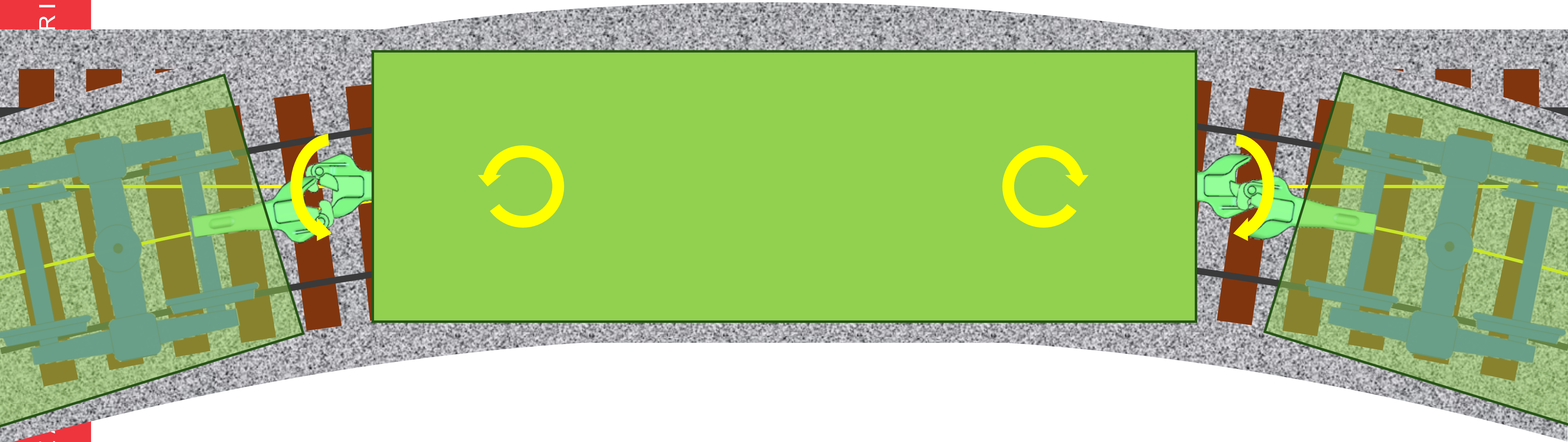
E/L 32% max

*used to be 38% until 2003

DALLAS, TX

Geometry of Curving

- Wordle:
- ✓ Truck Rotation
 - ✓ Coupler Swing
 - ✓ Knuckling



Car Design Considerations

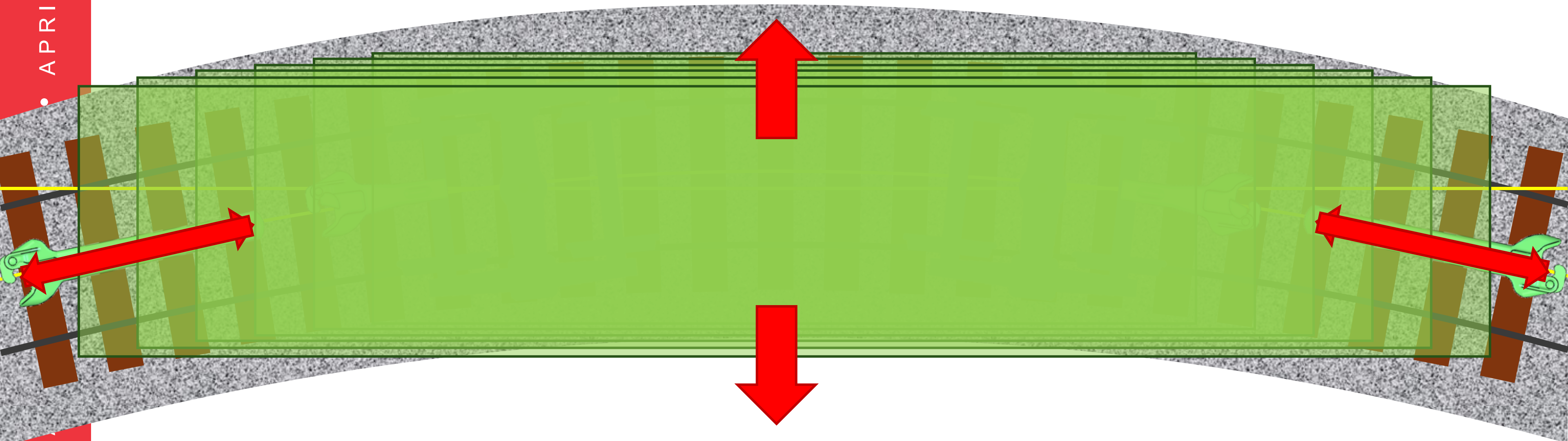


Wordle:

- ✓ Coupled Length
- ✓ Curve Negotiation
- ✓ Coupler Swing

Jackknife

Stringline

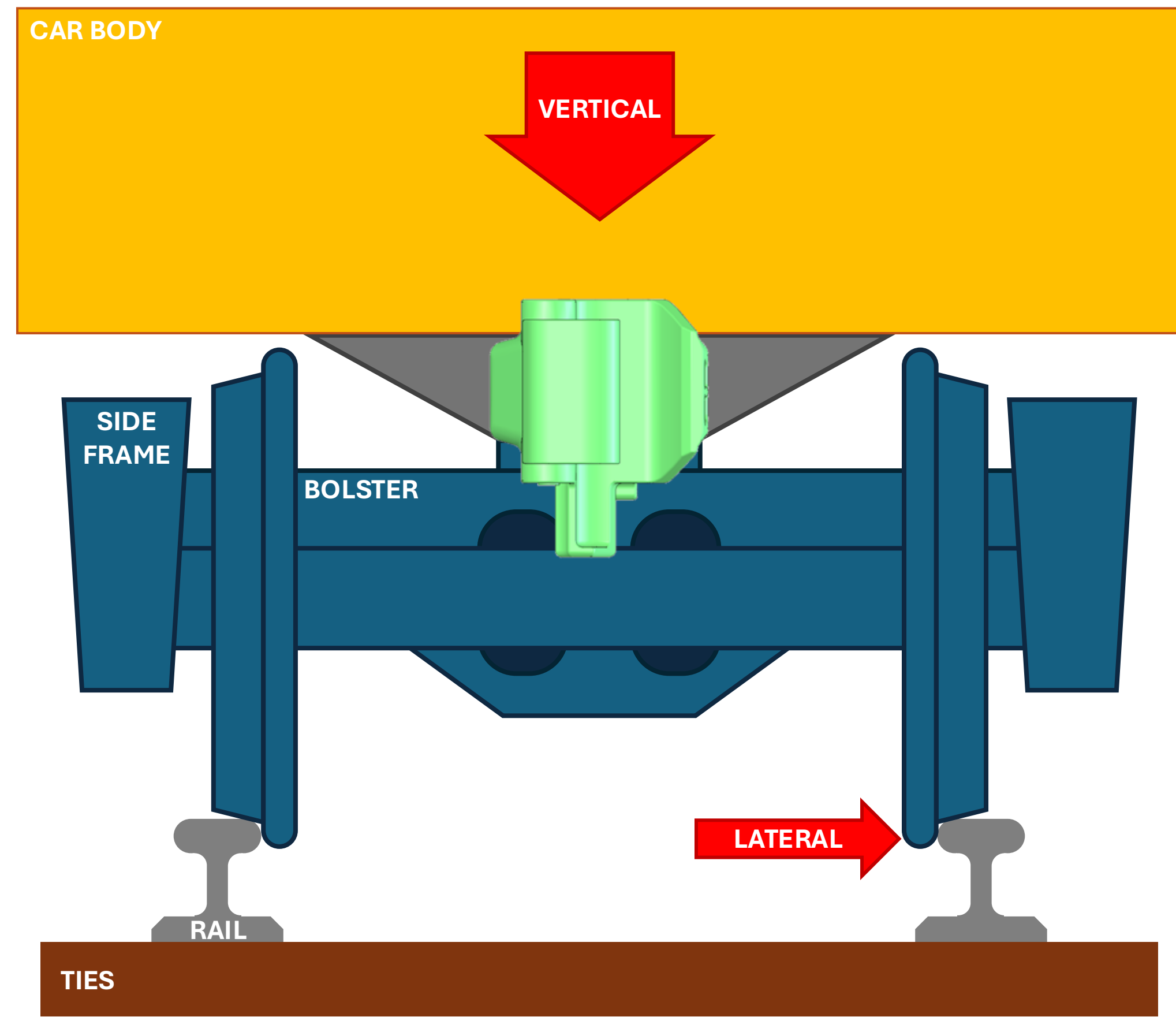




L/V Ratio



- Wordle:
- ✓ L/V Ratio
 - ✓ Flange Climb
 - ✓ Nadal Limit

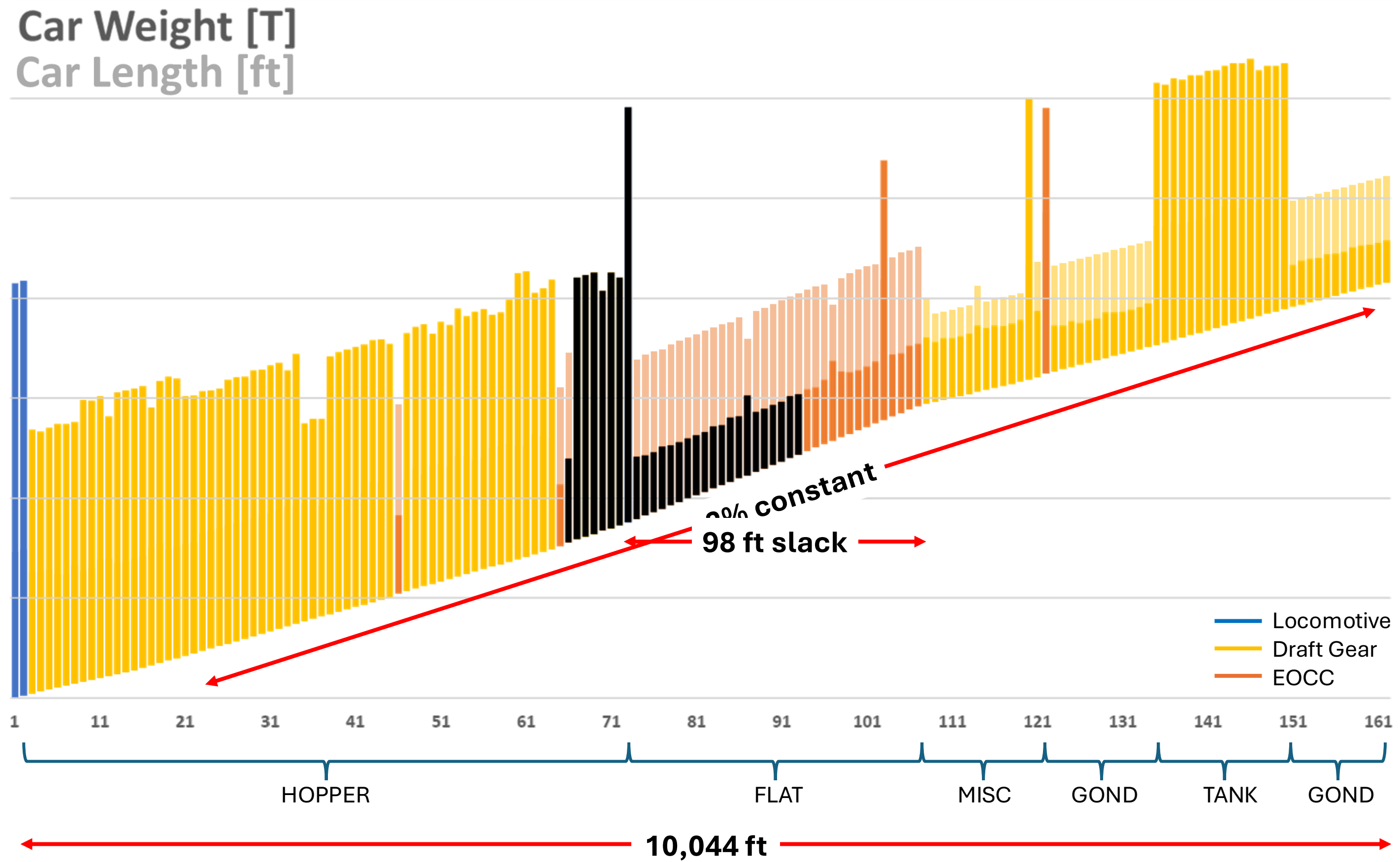


$$\frac{Lateral}{Vertical} = 0.82 MAX$$





Derailment Scenario – Consist



DALLAS, TX



Derailment Scenario



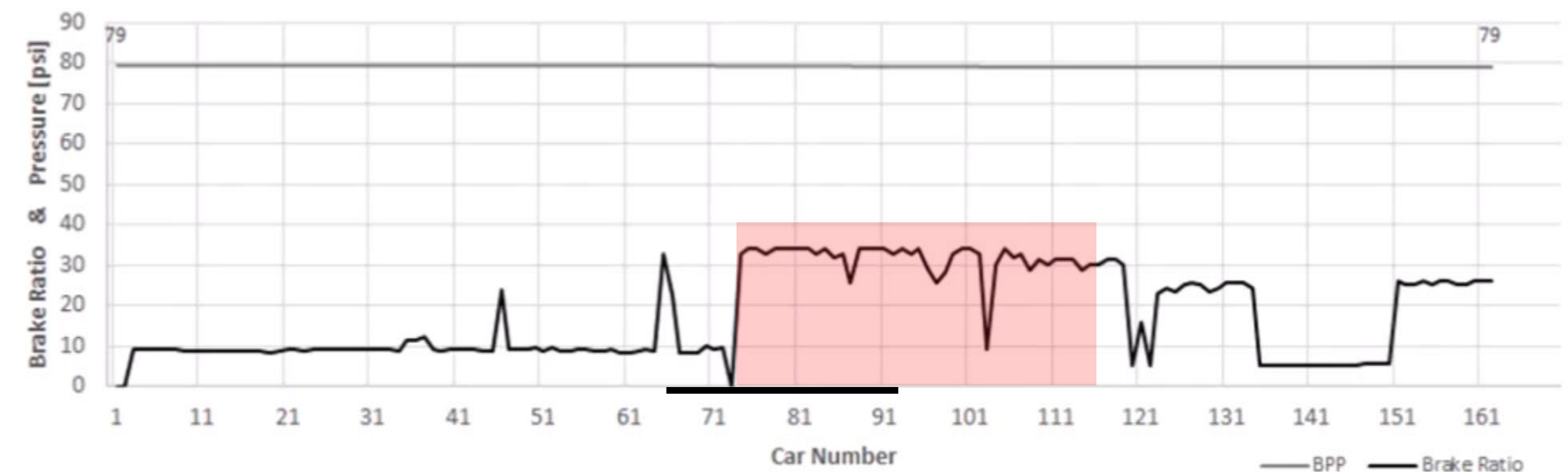
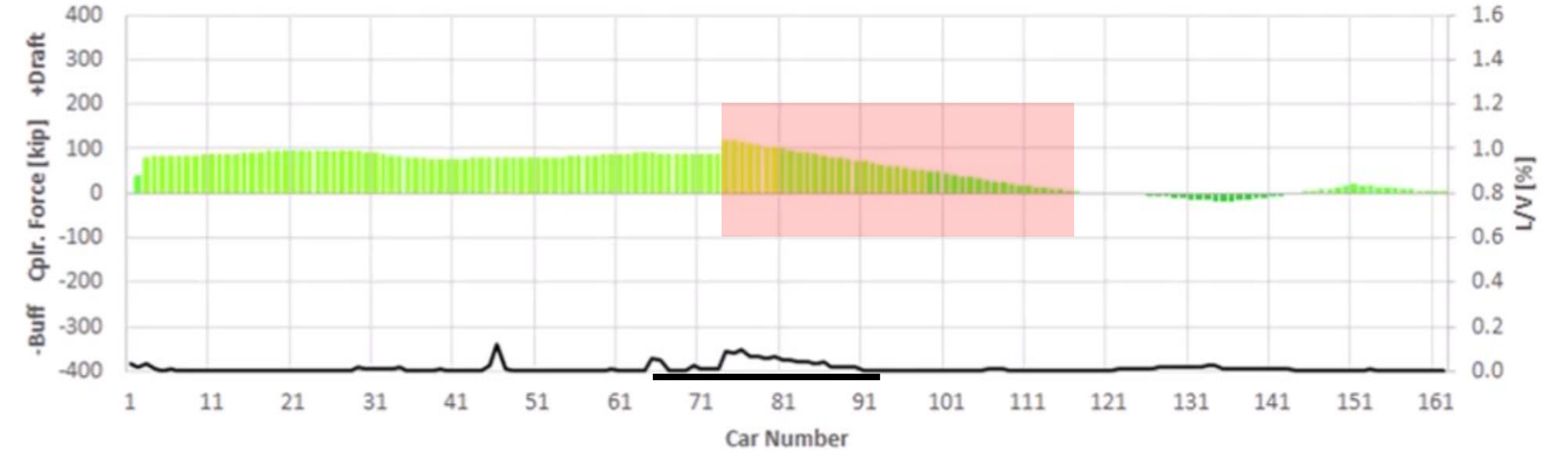
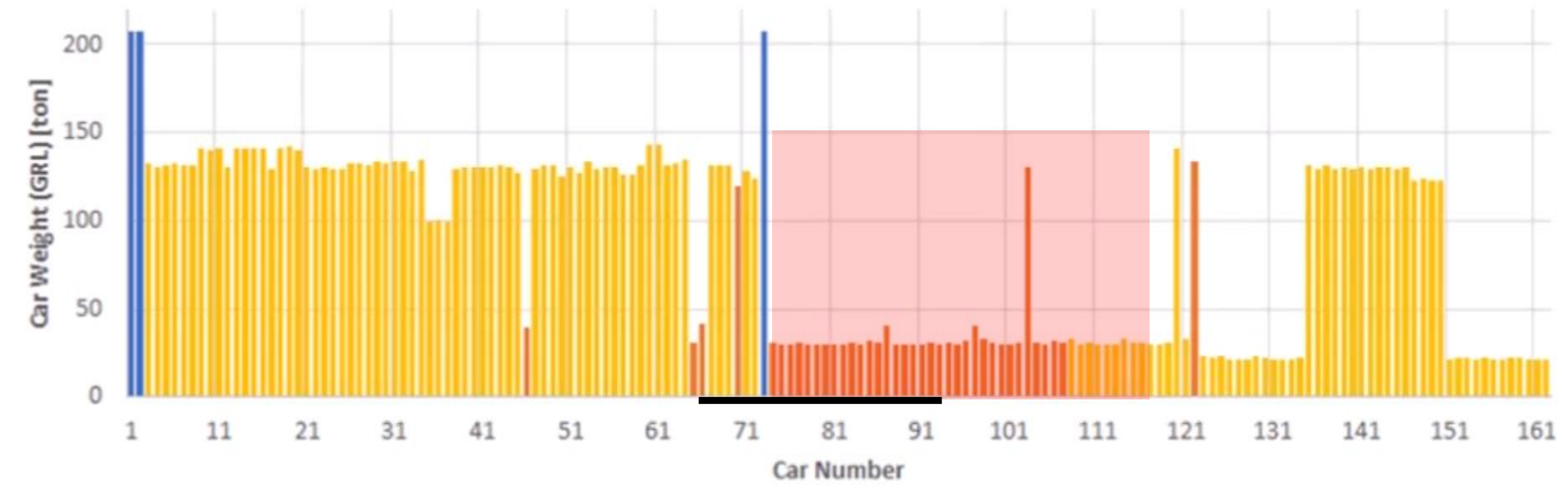
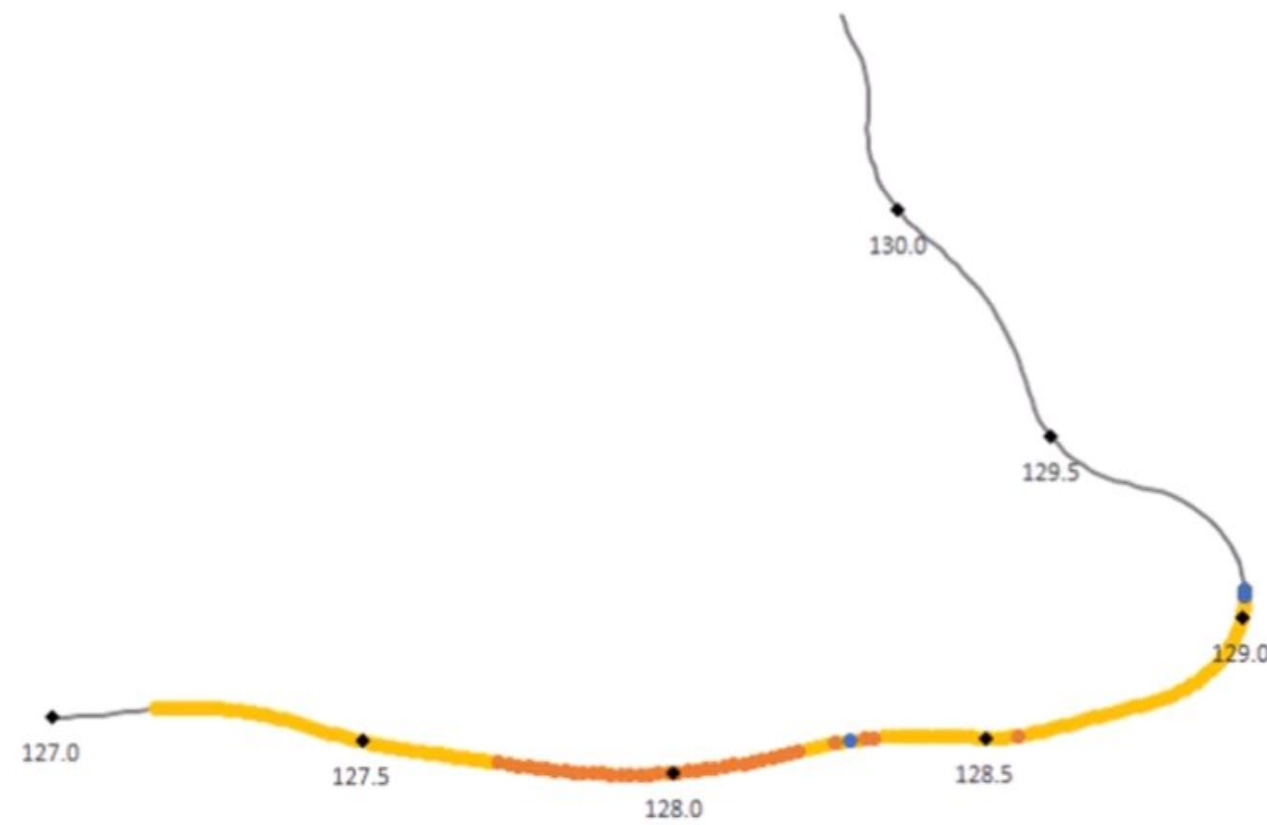
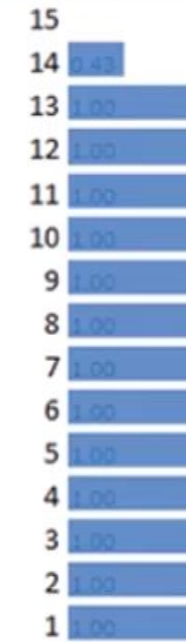
100 seconds

Draft forces at head end

E/L higher brake ratio, weight behind

Transition to buff through empty string

Notch	3
MPH	14.4



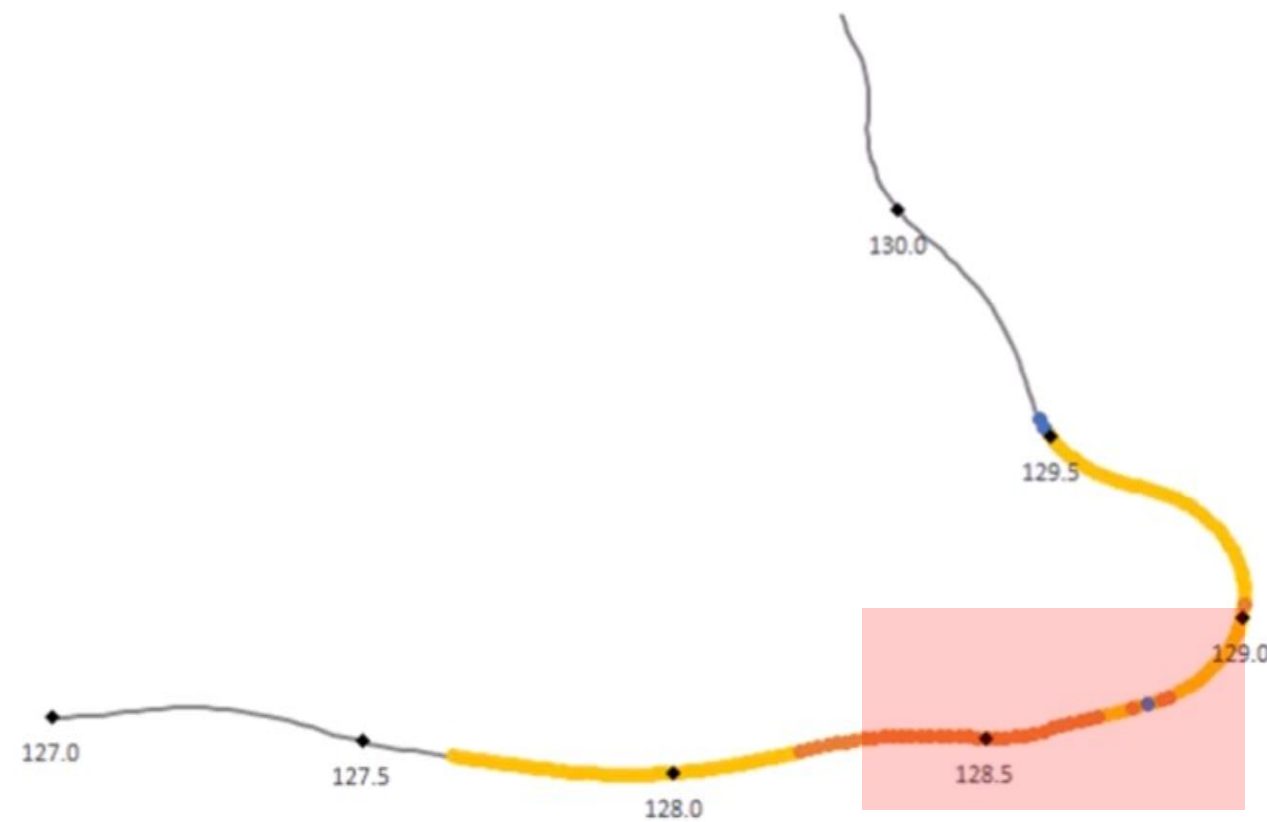


Derailment Scenario

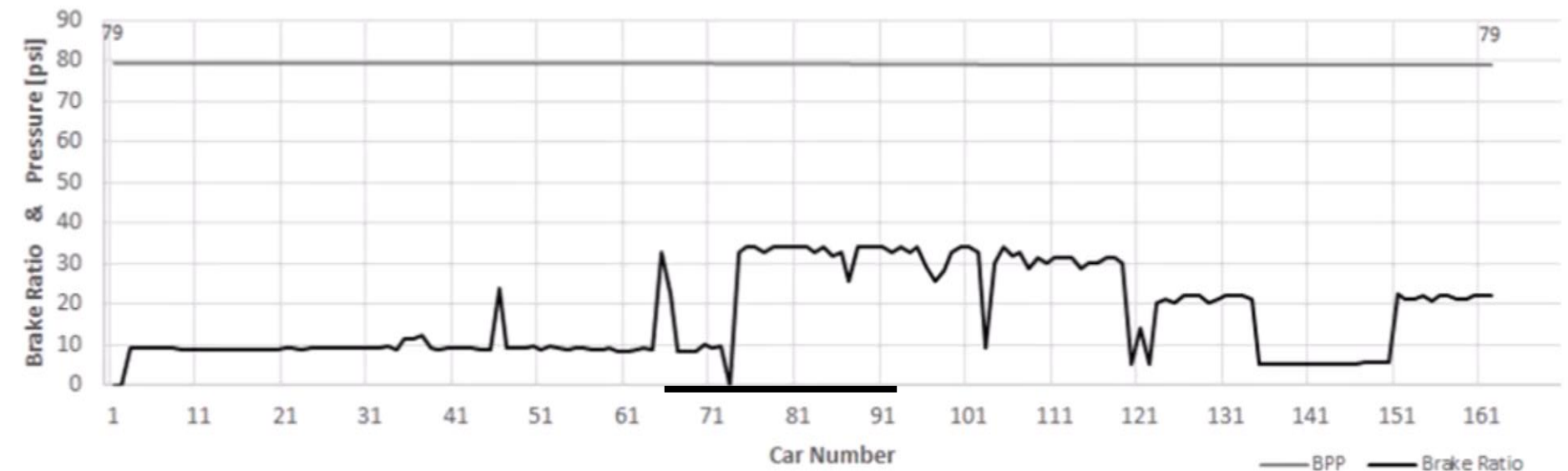
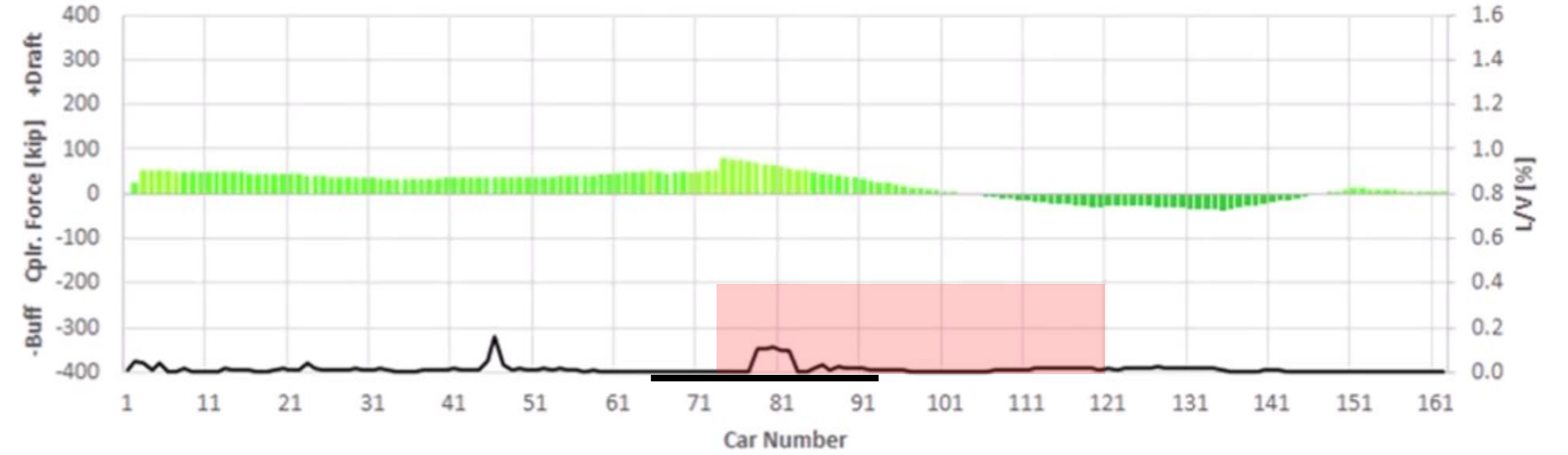
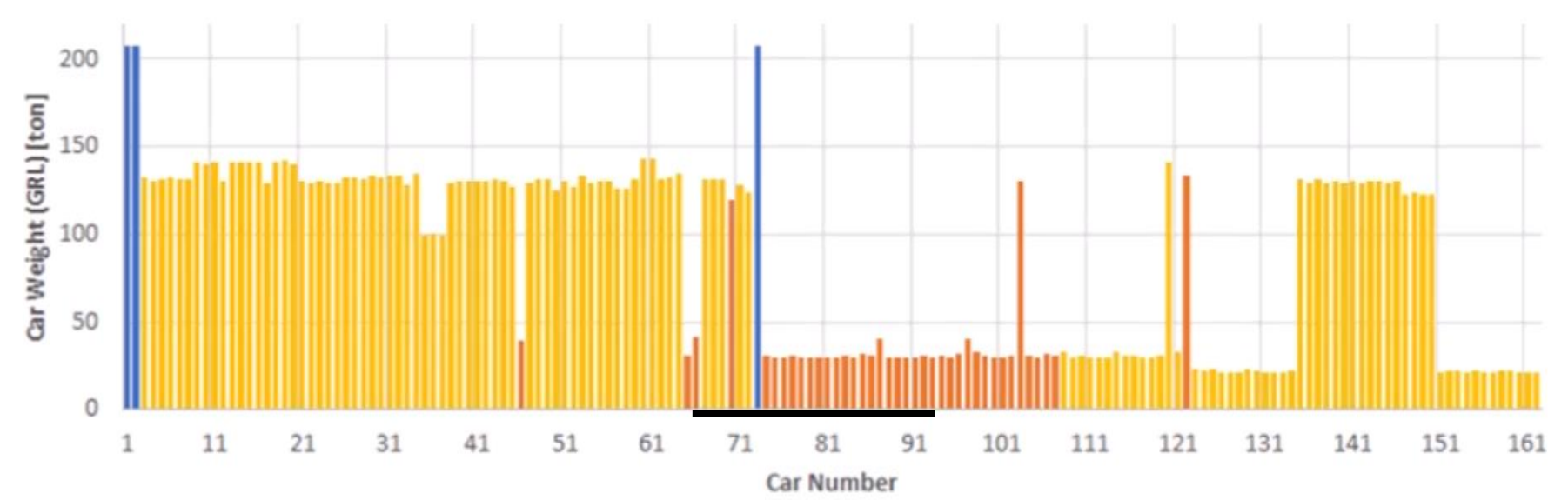
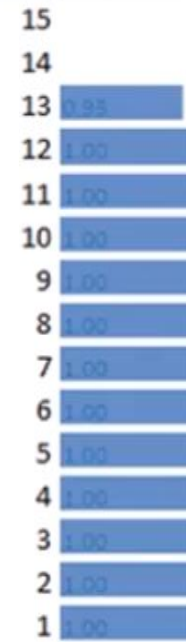


77 seconds

L/V “caterpillar” at curve entry



Notch	3
MPH	13.9



DALLAS, TX



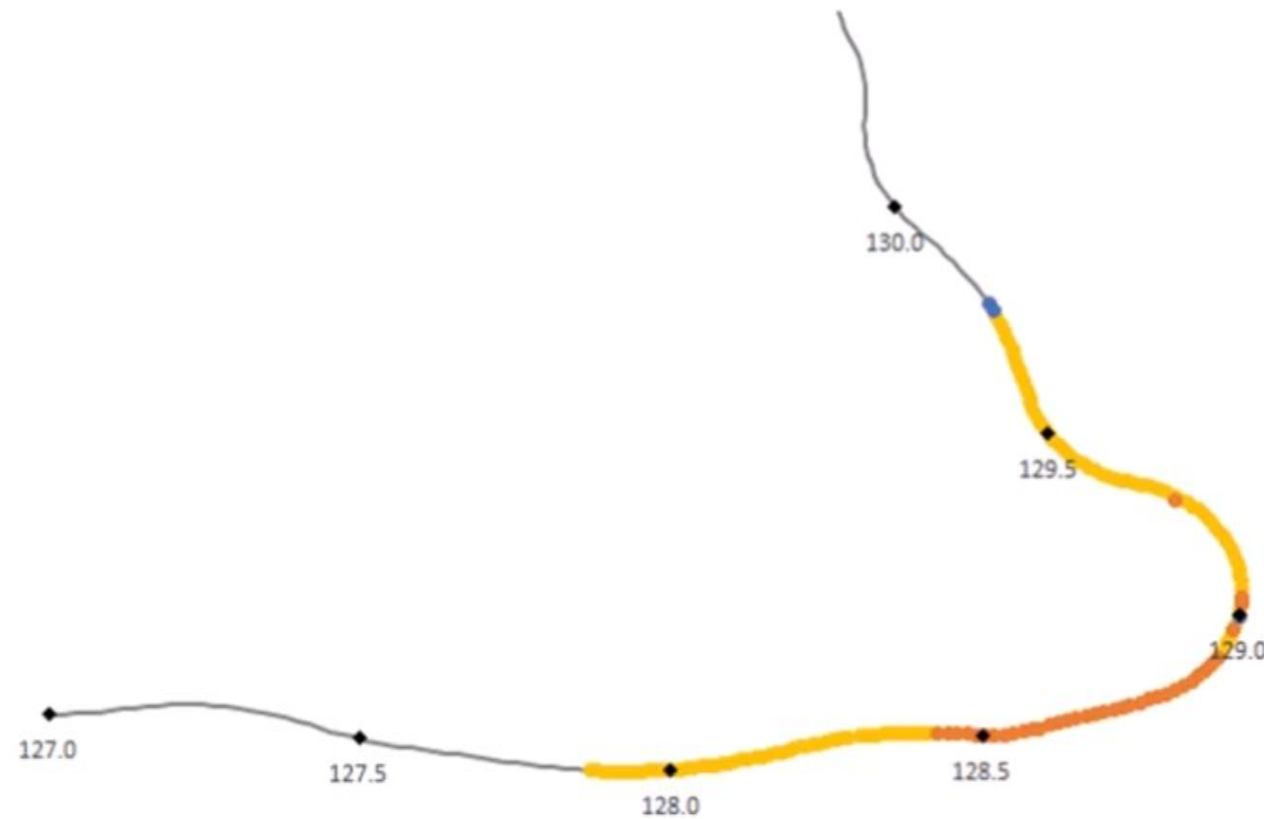
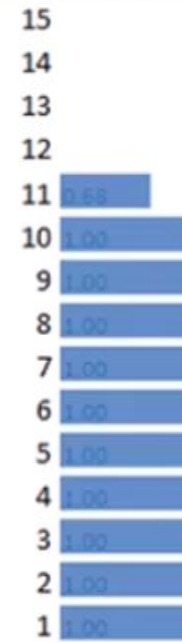
Derailment Scenario



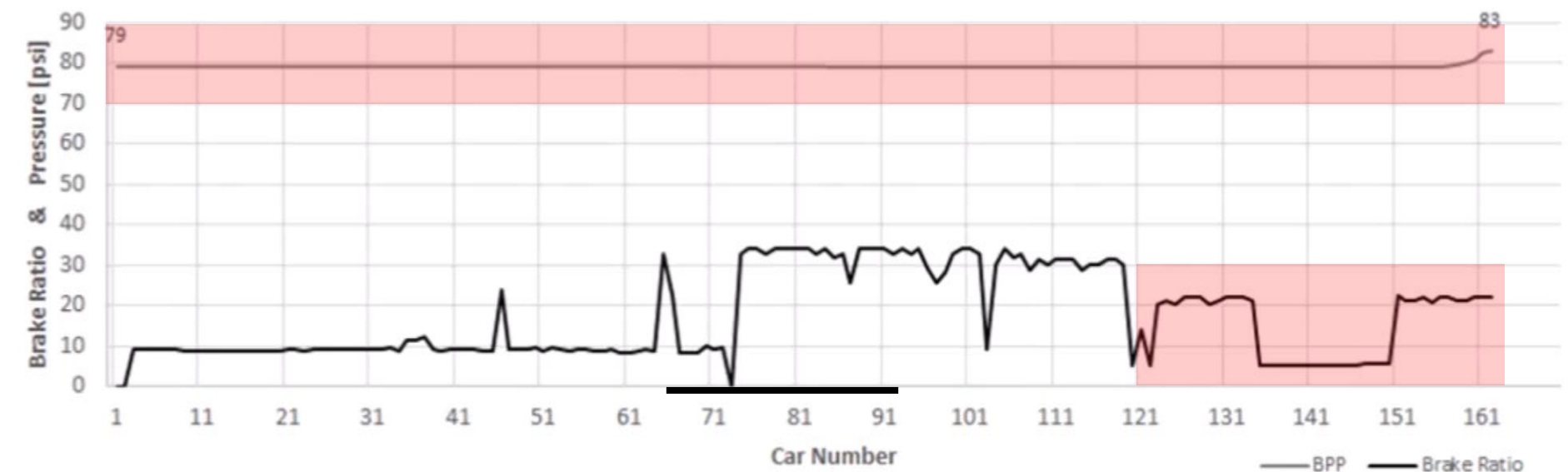
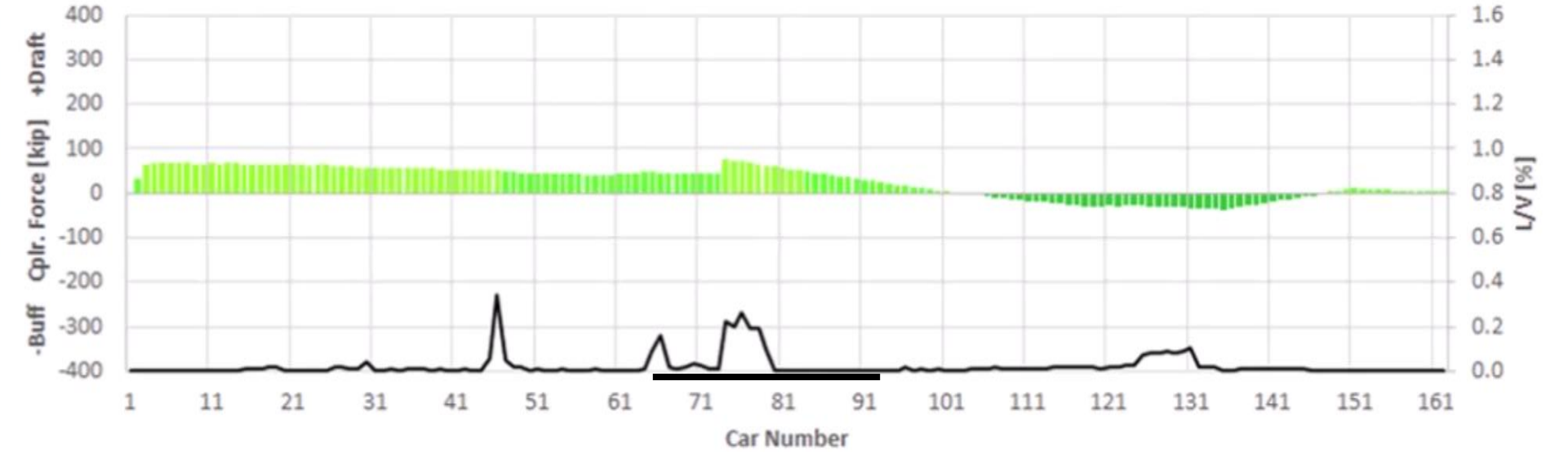
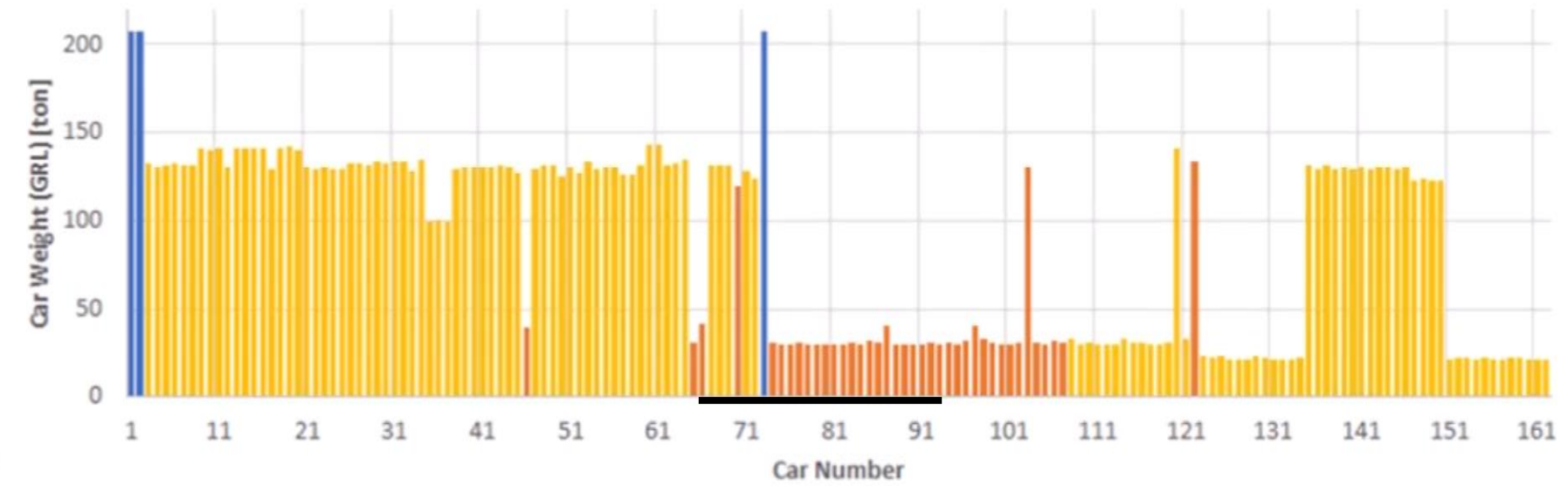
18 seconds

Unintended brake release from rear
Brake pipe pressure reaches head end

Notch	3
MPH	11.7



177



DALLAS, TX

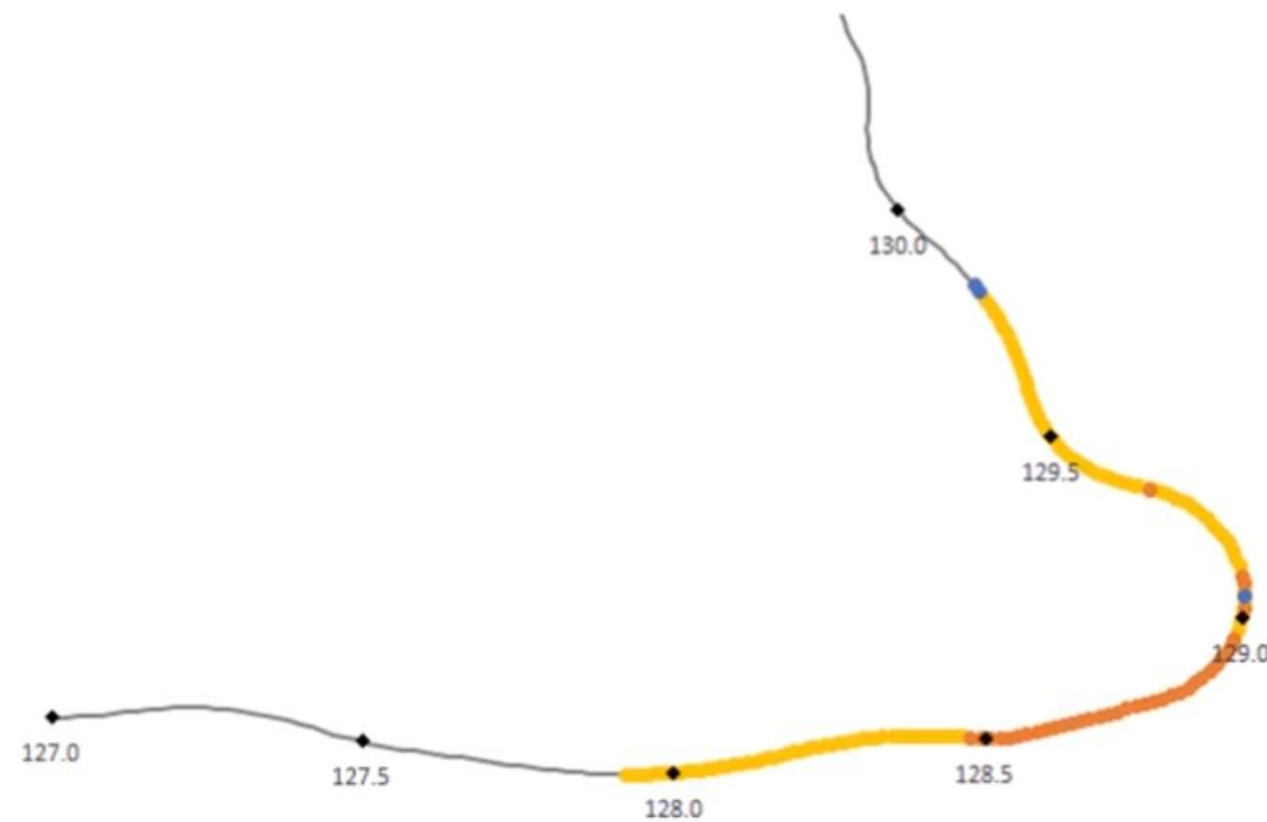


Derailment Scenario



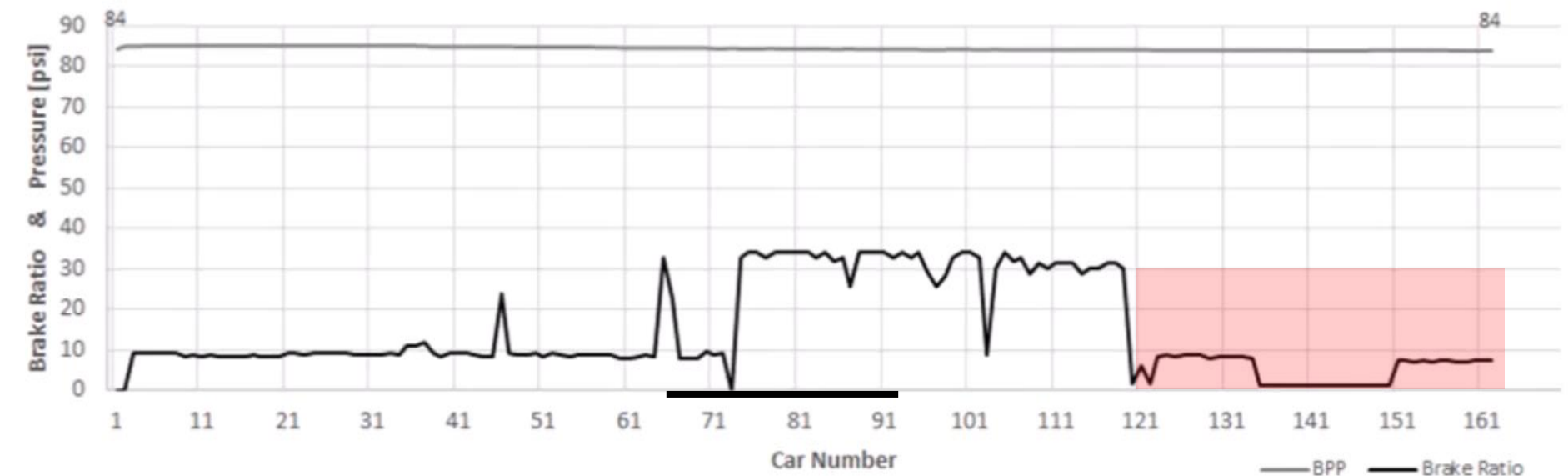
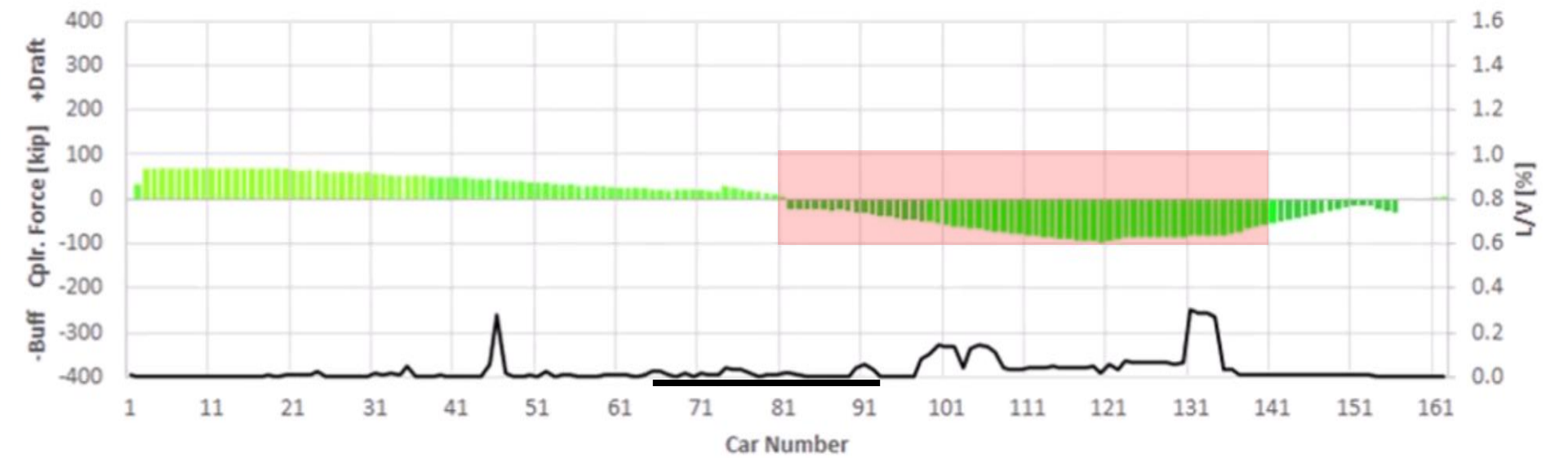
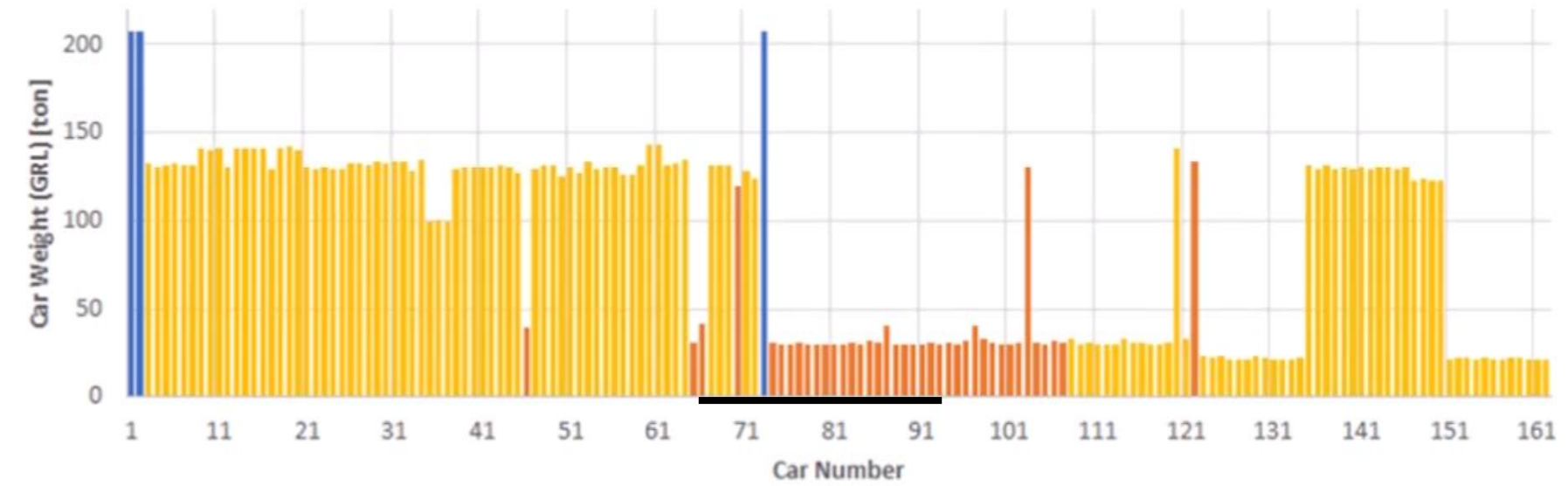
40 seconds

Brakes fully released at rear
Consist bunching in middle
Speed increasing



Notch	3
MPH	11.5
15	
14	
13	
12	
11	0.45
10	1.00
9	1.00
8	1.00
7	1.00
6	1.00
5	1.00
4	1.00
3	1.00
2	1.00
1	1.00

195



DALLAS, TX



Derailment Scenario



24 seconds

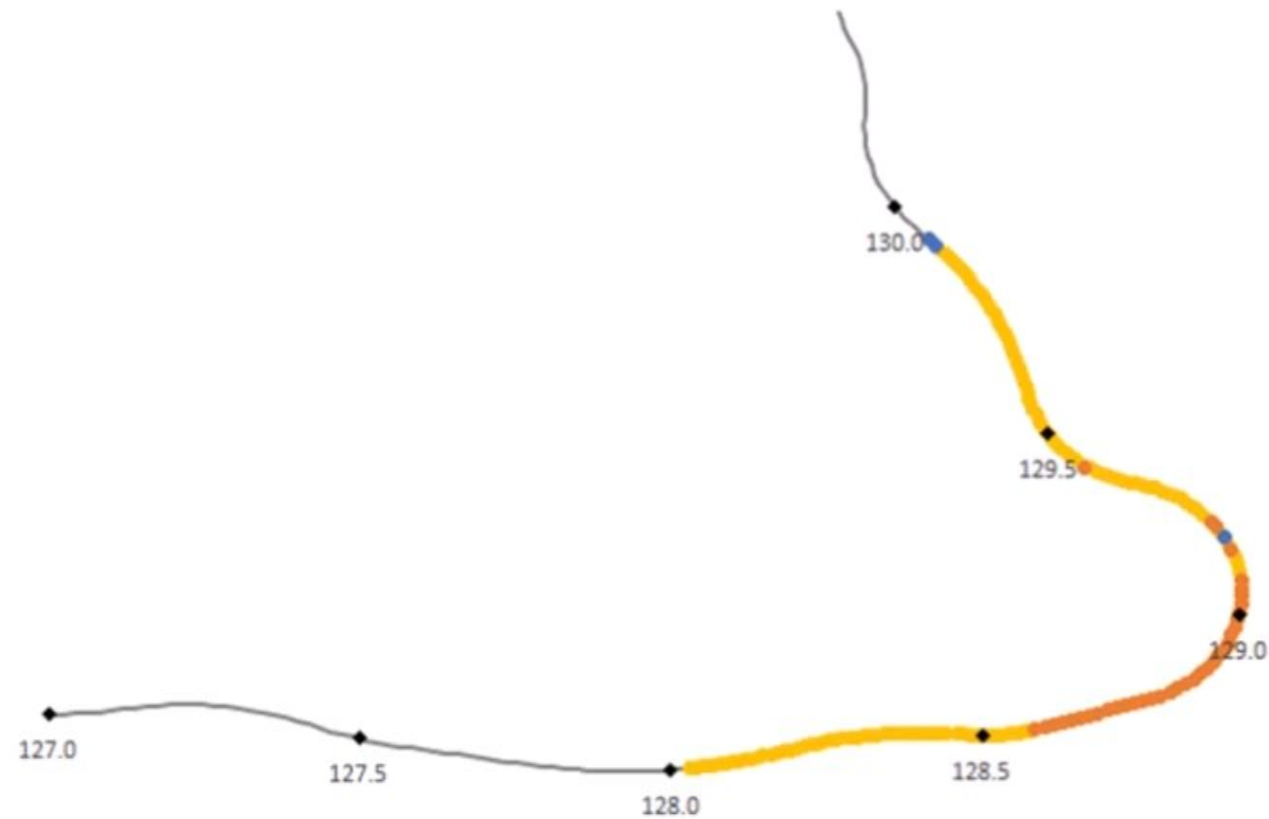
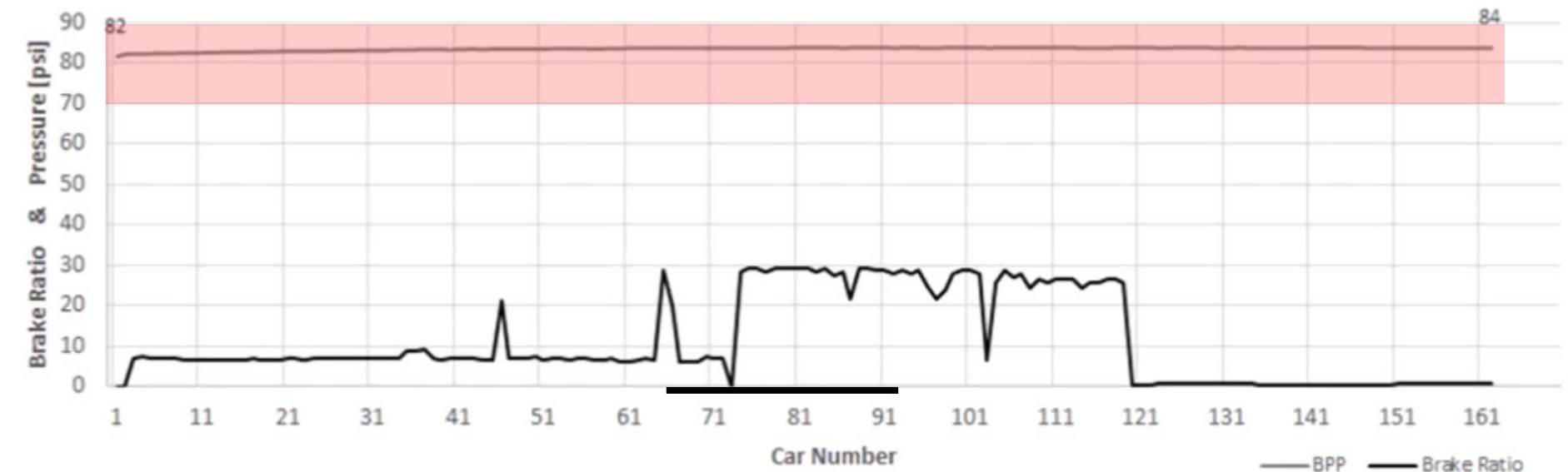
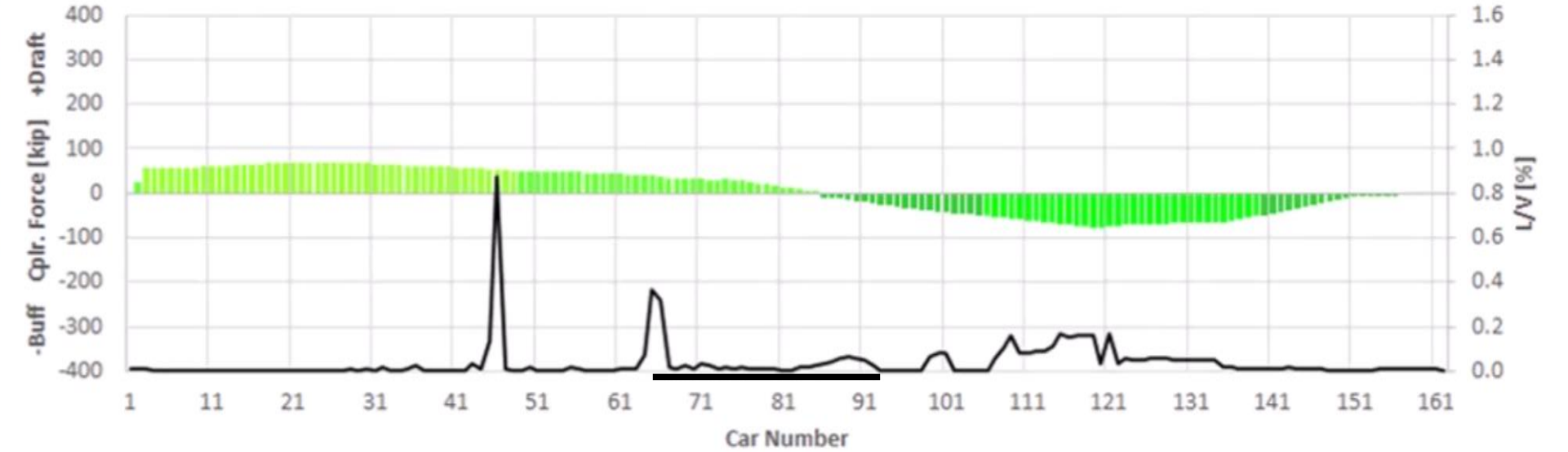
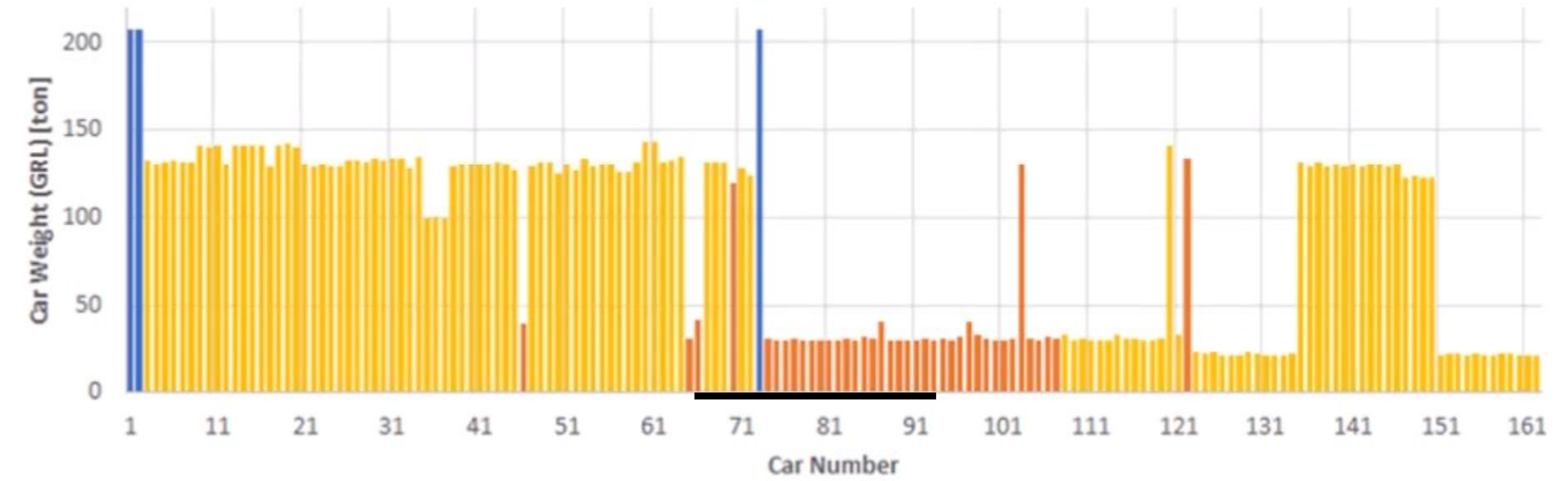
EMER application

Switch from draft to buff

Notch	0
MPH	13.9



232



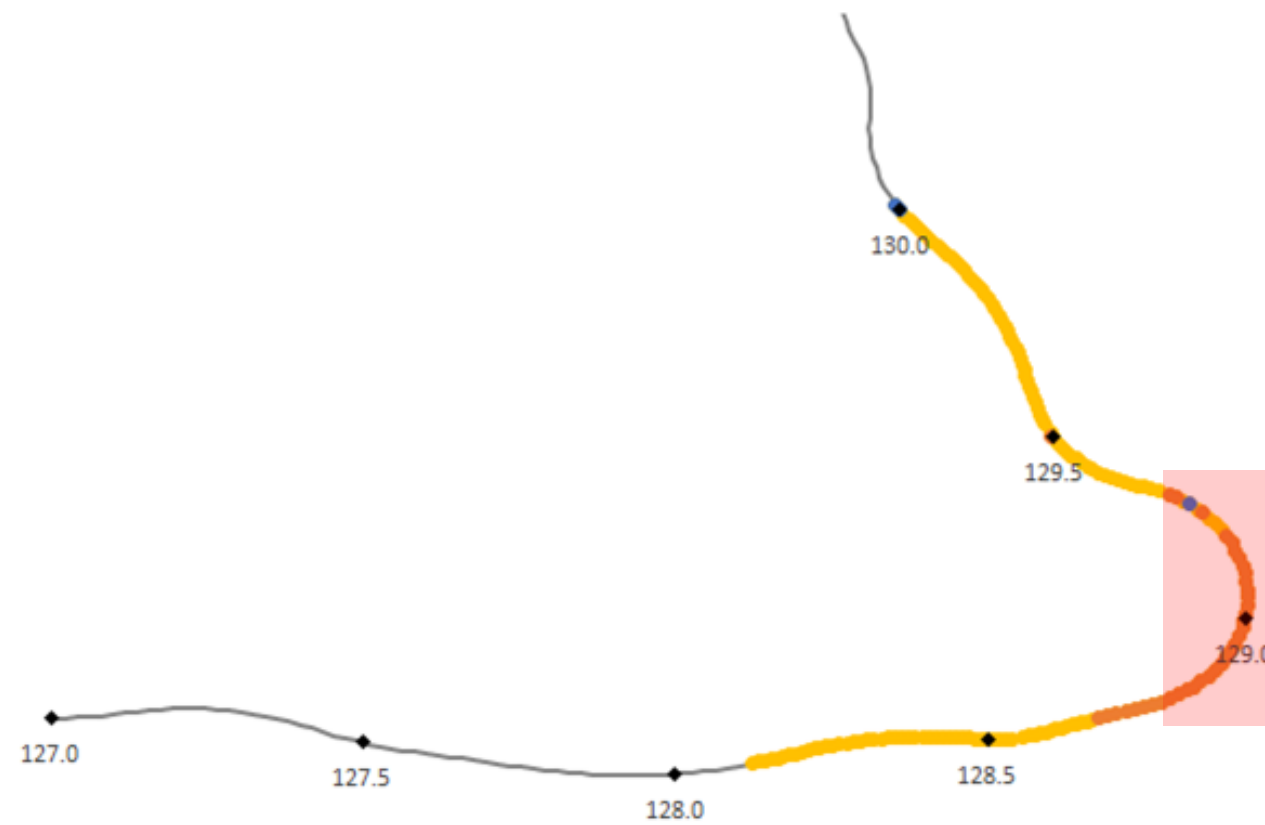
DALLAS, TX



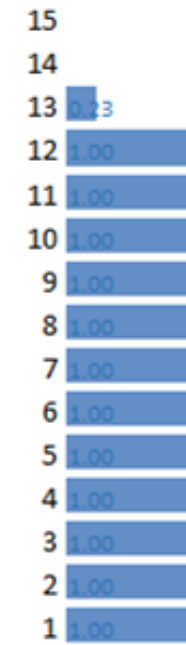
Derailment Scenario



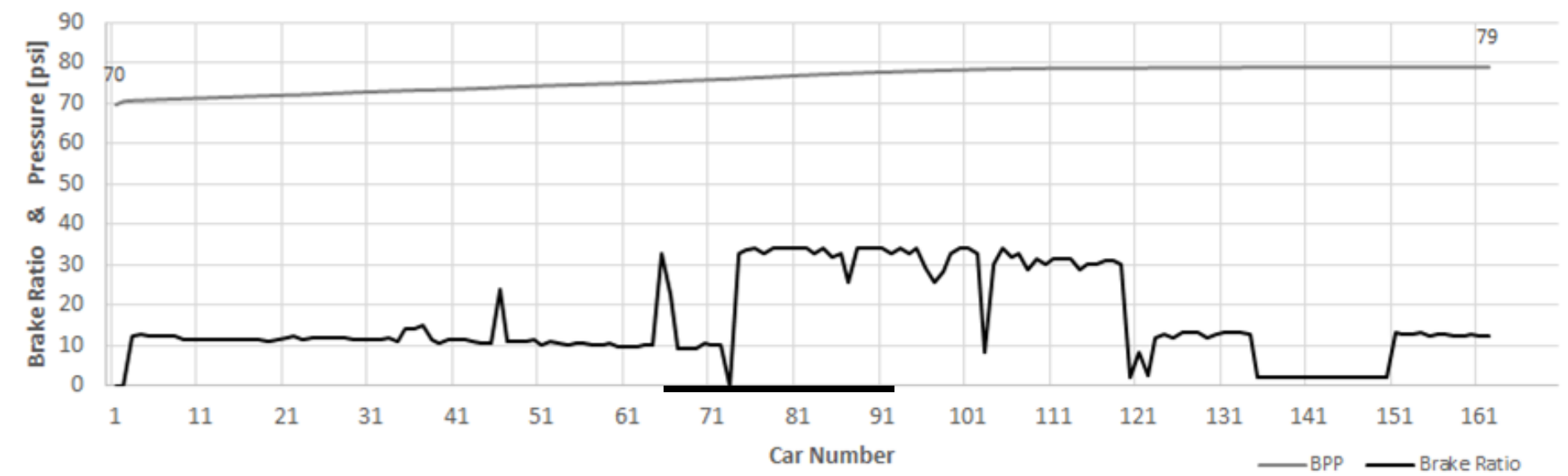
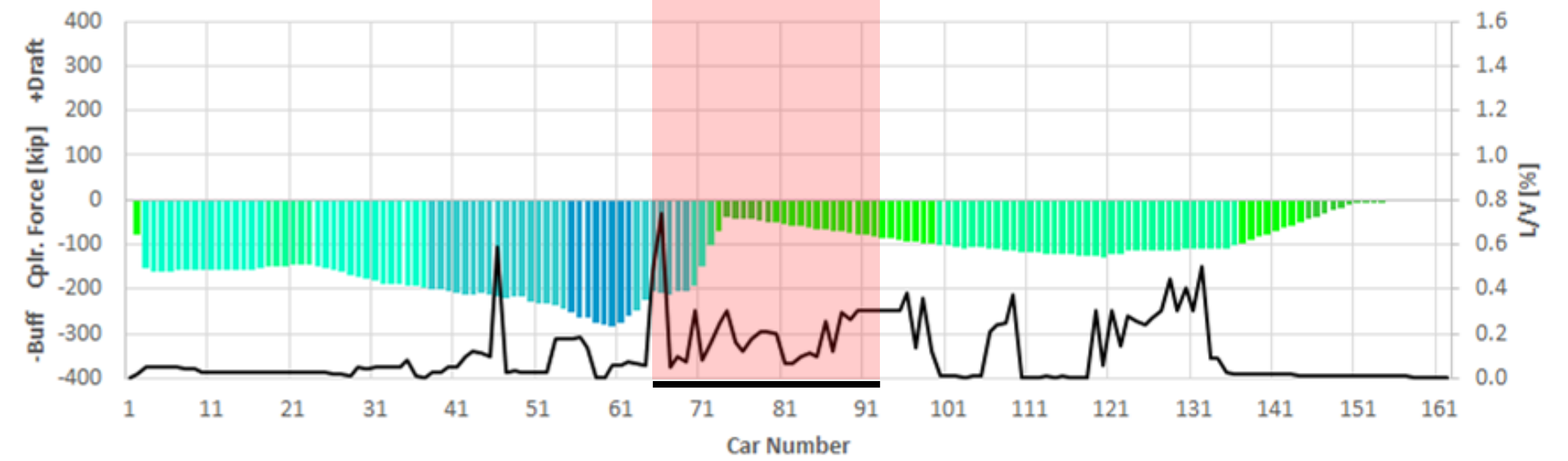
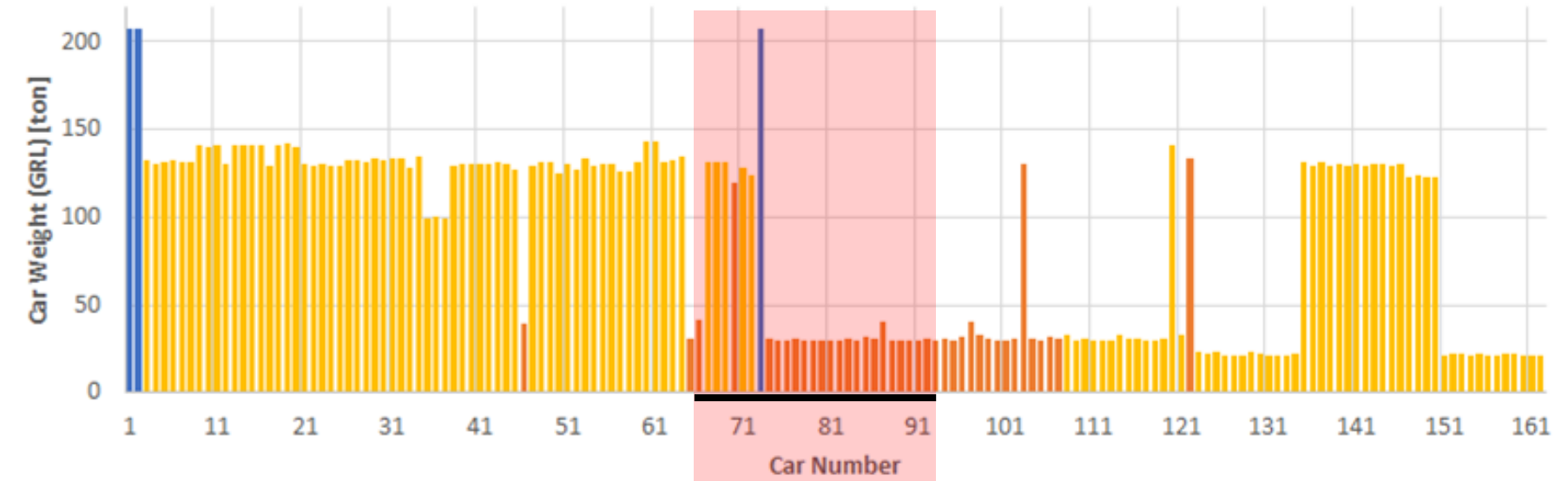
66-93 derail on curve
Full buff forces



Notch	-73
MPH	13.2



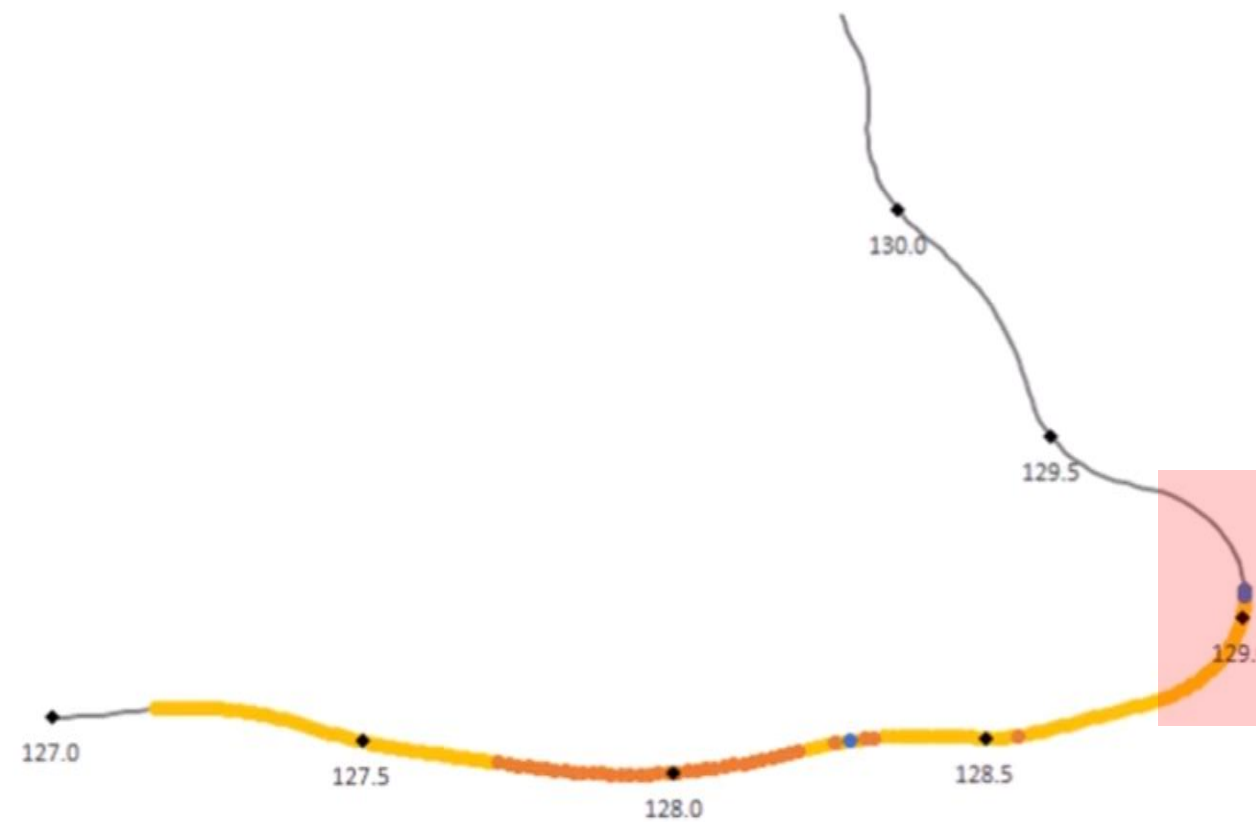
261





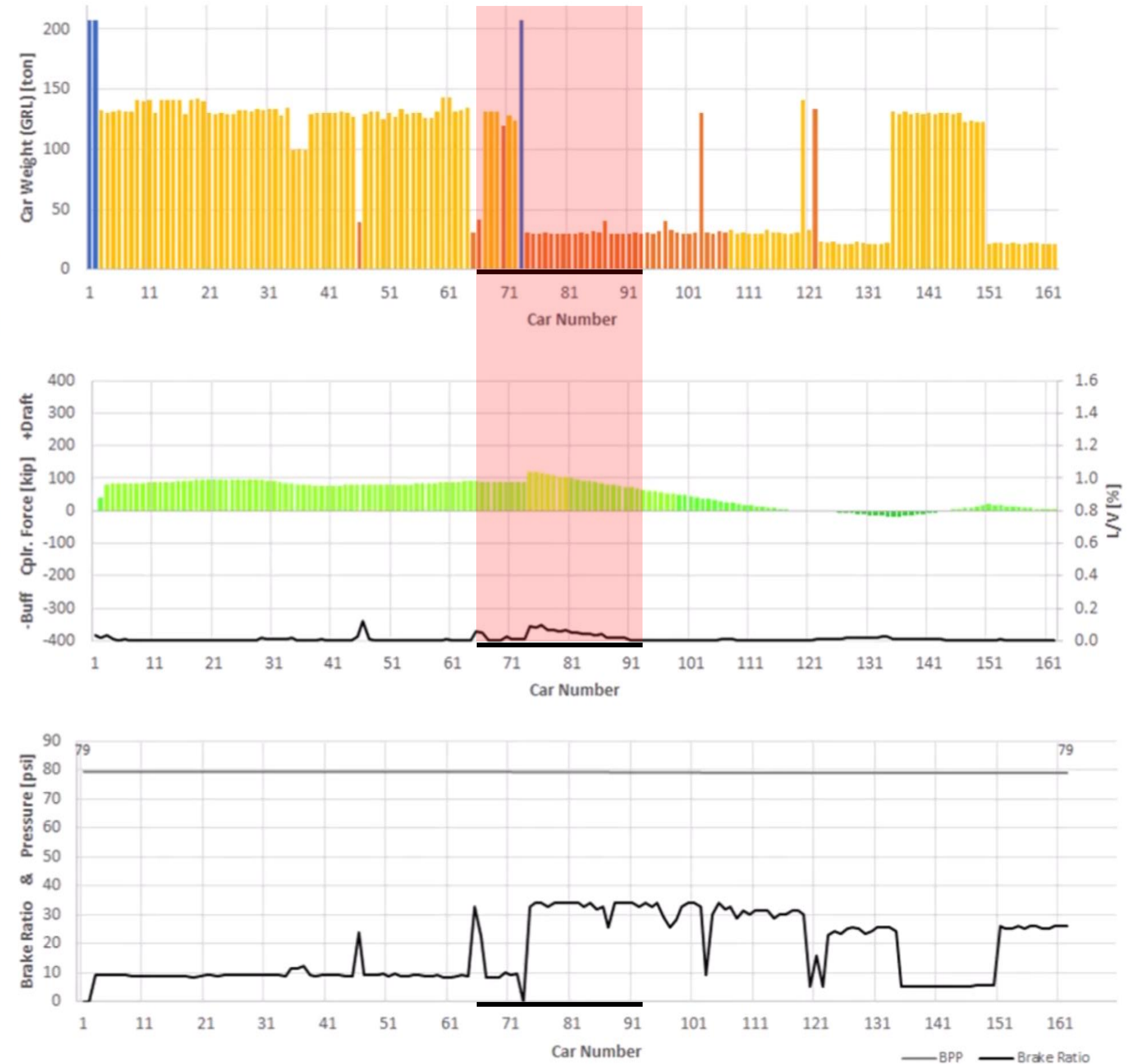
Derailment Scenario – Replay

66-93 derail on curve



Notch	3
MPH	14.4

15	1.00
14	1.00
13	1.00
12	1.00
11	1.00
10	1.00
9	1.00
8	1.00
7	1.00
6	1.00
5	1.00
4	1.00
3	1.00
2	1.00
1	1.00





Prevention



Train makeup
Draft system selection
Valve testing

